

MA 204N- 2-0-2-Syllabus:

Interpolation by polynomials, divided differences, error of the interpolating polynomial.

Solution of a system of linear equations, Cholesky's method, Gauss- Seidel methods, partial pivoting, row echelon form, norms, ill- conditioning. Eigen-value problem, power method.

Solution of a nonlinear equation, bisection and secant methods, Newton's method, rate of convergence, solution of a system of nonlinear equations

Numerical integration, composite rules, error formulae.

Numerical solution of ordinary differential equations, Euler and Runge-Kutta methods, multi-step methods, predictor-corrector methods, order of convergence.

Finite difference methods, numerical solutions of elliptic, parabolic, and hyperbolic partial differential equations

Suggested Readings:

1- S. S. Sastry, Introductory Methods of Numerical Analysis, PHI Learning, ISBN-978-81-203-4592-8, 2012.

2- E. Kreyszig, Advanced Engineering Mathematics, John Wiley and Sons, 2020, ISBN: 9781119455929.

3- S. D. Conte and Carl de Boor, Elementary Numerical Analysis - An Algorithmic Approach, SIAM, 2018, ISBN: 9781611975208

4- B. Bradie, A Friendly Introduction to Numerical Analysis, Pearson Prentice Hall, 2007, ISBN: 8131709426

5- W. Cheney, D. Kincaid, Numerical Mathematics and Computing, Cengage Learning, 2020, ISBN: 9780357670842.

6- D. Watkinson, Fundamentals of Matrix Computations, Wiley Inter Science, 2010, ISBN: 9780470528334.

7*- Kendall E. Atkinson, An Introduction to Numerical Analysis, 2ed, Wiley Publications, 1988

These are some **partial, rough** notes prepared for MA-204 students at IIT Indore. **Mistakes and typos** are bound to be there. Students are suggested to read them carefully, and are encouraged to send their comments and suggestions.

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1 Introduction

There are many mathematical problems, which are formulated while solving problems from other sciences. Some of these problems can be solved by methods which we have learned in the course of Calculus, Differential Equations and Linear algebra etc. And we are happy to get exact solutions or some information which is showing the analytic behaviour of the solution. But in most of the practical purpose we can not get exact solutions by direct computations (may be because of round off errors) or the solutions in known compact forms may not exist or even the solution in compact forms is not sufficient and we want some numeric value as the solution. In this course our aim is to solve such mathematical problems using numerical methods.

- * -Direct Methods, Numerical Approximations, Other approximations

- * Why- Numerical Analysis

- The Mathematical Problem is known to have solution but does not have any method to obtain exact solution

- The Mathematical problem is too difficult to be solved exactly

- There is no adequate information about the necessary inputs for the Mathematical Problem and therefore not possible to use the available exact methods

- * Who uses-

- Mathematicians, statisticians, Physicists, Chemists, Biologists, Engineers etc

- *why we use Numerical Methods

- Physical Experiments has its limitations like, cost, time, affordability

- because of development of computer technology, scientific computing is efficient and cheap

- readymade softwares - Matlab, Mathematica, Maple

- *Why theory involved in Numerical Methods

- helps to develop new Numerical Methods,

- choosing an appropriate among all available methods to produce more accurate results in lesser time

- to examine what goes wrong when the results are not as expected

Will learn some mathematical tools to do error analysis, arithmetic error

- System of linear algebraic equations, eigenvalue problems,

- Nonlinear system of algebraic equations, roots of polynomial and transcendental equations

- Interpolating polynomials, consider a few data points of a function and construct a polynomial satisfying the data, use it to predict the value in between the data

differentiation and integration

- System of linear equations, we may use some particular methods

for any typical problem there are typical methods to give some numbers, that can be used as solution,

therefore it is important to keep a possible track of errors in finding such approximate numerical solution

- make any meaning, or they are approximate solutions with some unknown errors, can we make these errors sufficiently small,

therefore we need to know the sources of errors, do they get magnified, (and in that case they spoil the solution) final outcome is how accurate and how reliable

Error = true value - approximate value

Absolute error = —Error—

relative error = $\frac{|\text{error}|}{|\text{true Value}|}$

is a good indicator

Errors in Computation (R.E.):

Round off error (RE) in a number: It is a quantity R which must be added to the finite representation of the number in order to make it the true representation of that number

Consider $f13$, with keeping four significant digits, then

$\frac{1}{3} = 0.3333$ and $R = 0.00003333.....$

rational numbers and roots of positive numbers

Round of error is not under the control of the user

Truncation error (TE): It is a quantity T which must be added to the representation of a quantity in order to make this quantity exactly equal to the number. It arises due to use of approximate formulas, usually based on Taylor series

$$f(x) = f(y) + (x - y)f'(y) + \dots + \frac{(x-y)^n}{n!}f^n(y) + R$$

where R is remainder

$$R = \frac{(x-y)^{n+1}}{(n+1)!}f^{n+1}(\xi)$$

Example 1.1. $\sin x = x - \frac{x^3}{3!} + R$

$R = \frac{x^4}{4!}\sin(\xi)$ for some ξ in between 0 and x , therefore

$$|TE| \leq \frac{|x|^4}{4!}$$

Truncation error is under the control of the user. If we need better approximation, we can take more terms in the Taylor series.

Total error in computation = roundoff error + TE + other errors

may include –

-the "machine epsilon" error for converting a number to its binary form

-subtraction of two nearly equal numbers

Let $x^* \sim x$, $y^* \sim y$, $z^* = x^* - y^*$

$x^* = 0.264593 \times 10^1$, $y^* = 0.264162 \times 10^1$, correct to six significant digits

RE in x and $x - y$

$$\text{RE } x = \frac{0.5 \times 10^{-5}}{0.265493 \times 10^1} \sim 1.89 \times 10^{-6}$$

$$\text{RE } x - y = \frac{0.5 \times 10^{-5} + 0.5 \times 10^{-5}}{0.0000431 \times 10^1} \sim 2.32 \times 10^{-3}$$

which is very large compared to RE x ,

If the value of z^* is used for further computations, errors may increase

Therefore, when two nearly equal numbers are subtracted, there is always a loss of significant digits.

2- Division by a very small number

$$f(x) = \frac{1}{1-x^2}$$

To find $f(0.9)$

$$\text{Exact value} = \frac{1}{1-0.81} = \frac{1}{0.19} = 0.526316 \times 10^1$$

If we take $x \sim x^* = 0.900005$ with error $= 0.5 \times 10^{-5}$, then

$$f(x^*) = 0.526341 \times 10^1$$

therefore error in sixth place has produced an error in fifth place

therefore accuracy decreases further

What is the cumulative effect of all these errors in a large computation

"If the growth of errors is under control, (bounded) then the method gives stable results, otherwise, numerical instability occurs

Numerical instability can occur because of

-Inherent instability, - induced instability

Roots of $a_0x^2 + a_1x + a_2 = 0$

$$x = \frac{1}{2a_0}[-a_1 \pm \sqrt{a_1^2 - 4a_0a_2}]$$

operational count are

m-3,d-1,a/s-2,sqrt -1

2 Solution of a system of linear equations

Here we aim to solve the following system of linear equations.

$$\begin{array}{cccccccccccc}
 a_{11}x_1 & a_{12}x_2 & \dots & \dots & \dots & a_{1i}x_i & \dots & \dots & \dots & a_{1n}x_n & = & b_1 \\
 a_{21}x_1 & a_{22}x_2 & \dots & \dots & \dots & a_{2i}x_i & \dots & \dots & \dots & a_{2n}x_n & = & b_2 \\
 \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\
 \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\
 \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\
 a_{i1}x_1 & a_{i2}x_2 & \dots & \dots & \dots & a_{ii}x_i & \dots & \dots & \dots & a_{in}x_n & = & b_i \\
 \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\
 \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\
 \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\
 a_{n1}x_1 & a_{n2}x_2 & \dots & \dots & \dots & a_{ni}x_i & \dots & \dots & \dots & a_{nn}x_n & = & b_n
 \end{array} \tag{2.1}$$

In this system of linear equations $a'_{ij}s, i, j = 1, 2, \dots, n$ are known coefficients, $b'_i s, i = 1, 2, \dots, n$ are known values and $x'_i s, i = 1, 2, \dots, n$ are unknowns to be determined. This system can also be written in matrix form as

$$Ax = b, \tag{2.2}$$

where A is $n \times n$ matrix and x and b are column vectors of order $n \times 1$ of unknowns and known values respectively. If $b = 0$, then (2.2) is called homogeneous system of equations. For the solution of the system of linear equations (2.2) we recall an important theorem from linear algebra.

Theorem 2.1. *If A is real matrix of order $n \times n$, then the following statements are equivalent.*

- $Ax = 0$ has only trivial solution.
- For each b , $Ax = b$ has a solution.
- A is invertible.
- $\det(A) \neq 0$.

Remark 2.1. If any of the four equivalent conditions are satisfied in the above theorem, then one can find the solution of the system of linear equations by multiplying the inverse of the matrix A on left of both side of the equation $Ax = b$ to get

$$x = A^{-1}b. \tag{2.3}$$

And to find A^{-1} we know the standard method (learned in 10 + 2 standard) of finding the adjoint of the matrix A and $A^{-1} = \text{Adj}(A)/\det(A)$. One can also recall the Cramer's method of finding the solution.

Observation 2.1. Both the above method of finding the solution involve an intermediate step of finding the determinant of matrix A . But finding the determinant of a large matrix is not an easy task.

Question 2.1. Can we find the solution of the system of linear equations without finding the inverse of the coefficient matrix?

2.1. Direct method for some special form of the coefficient matrix A

Here we will try to answer the question (2.1) positively if the coefficient matrix is of some easy form such that the solution can be obtained by direct computations.

1. **Diagonal case:** $A = D$

Since matrix A is assumed to be nonsingular, $\det(A) = \prod_{i=1}^n a_{ii} \neq 0$. And we have

$$\begin{bmatrix} a_{11} & 0 & \dots & \dots & 0 & \dots & \dots & 0 & 0 \\ 0 & a_{22} & \dots & \dots & 0 & \dots & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \dots & a_{ii} & \dots & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \dots & 0 & \dots & \dots & a_{n-1,n-1} & 0 \\ 0 & 0 & \dots & \dots & 0 & \dots & \dots & 0 & a_{nn} \end{bmatrix} \times \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ \dots \\ x_i \\ \dots \\ \dots \\ x_{n-1} \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \dots \\ \dots \\ b_i \\ \dots \\ \dots \\ b_{n-1} \\ b_n \end{bmatrix}.$$

The solution is obvious in this case and can be written as

$$x_{ii} = \frac{b_i}{a_{ii}}, \quad i = 1, 2, \dots, n. \quad (2.4)$$

Note that the only n divisions are required as computer operations.

2. **Lower triangular case:** $A = LT$

We have the following matrix equation.

$$\begin{bmatrix} a_{11} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \dots & \dots & \dots & 0 & 0 & 0 & 0 & 0 & 0 \\ \dots & \dots & \dots & \dots & 0 & 0 & 0 & 0 & 0 \\ a_{i1} & a_{i2} & \dots & \dots & a_{ii} & 0 & 0 & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & 0 & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & 0 & 0 \\ a_{n-1,1} & a_{n-1,2} & \dots & \dots & a_{n-1,i} & \dots & \dots & a_{n-1,n-1} & 0 \\ a_{n1} & a_{n2} & \dots & \dots & a_{ni} & \dots & \dots & a_{n,n-1} & a_{nn} \end{bmatrix} \times \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ \dots \\ x_i \\ \dots \\ \dots \\ x_{n-1} \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \dots \\ \dots \\ b_i \\ \dots \\ \dots \\ b_{n-1} \\ b_n \end{bmatrix}.$$

In this case also we assume the solution to exist and hence, $\det(A) = \prod_{i=1}^n a_{ii} \neq 0$. Here it is easy to compute x_1 but to compute x_2 , we need to substitute the value of x_1 in the second equation. Similarly to compute x_k we need to substitute the values of $x_1, x_2, \dots, x_{k-1}$ (which are already obtained) in the following equation

$$x_k = \frac{(b_k - \sum_{j=1}^{k-1} a_{kj}x_j)}{a_{kk}}, \quad k = 2, \dots, n. \quad (2.5)$$

Thus we need forward substitution of the entries so we call this method as forward substitution. For computation of x_k in the above equation we require $(k-1)$ multiplications, $(k-1)$ additions and one division. Thus the total number of computer operations are $\sum_{k=1}^n (k-1) + (k-1) + 1 = n^2 + 2n$.

3. **Upper triangular case:** $A = UT$

In this case we also have $a_{ii} \neq 0$. And we have

$$\begin{bmatrix} a_{11} & a_{12} & \dots & \dots & a_{1i} & \dots & \dots & a_{1,n-1} & a_{1n} \\ 0 & a_{22} & \dots & \dots & a_{2i} & \dots & \dots & a_{2,n-1} & a_{2n} \\ 0 & 0 & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & 0 & a_{ii} & \dots & \dots & a_{i,n-1} & a_{in} \\ 0 & 0 & 0 & 0 & 0 & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & 0 & 0 & 0 & \dots & \dots & \dots \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & a_{n-1,n-1} & a_{n-1,n} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & a_{nn} \end{bmatrix} \times \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ \dots \\ x_i \\ \dots \\ \dots \\ x_{n-1} \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \dots \\ \dots \\ b_i \\ \dots \\ \dots \\ b_{n-1} \\ b_n \end{bmatrix}.$$

But here the computation of x_n is easy compared to other unknowns. So we first compute x_n from the last equation and substitute its value in the second last equation to compute x_{n-1} . And to compute x_k we substitute the values of unknowns $x_n, \dots, x_{k+1}$ obtained from the last $n - k$ equations in the k_{th} equation as follows

$$x_k = \frac{(b_k - \sum_{j=k+1}^n a_{kj}x_j)}{a_{kk}}, \quad k = 2, \dots, n-1. \quad (2.6)$$

Because of substitution in the earlier equations this method is known as back substitution. And similar to forward substitution we need total $n^2 + 2n$ computer operations for the complete solution.

Remark 2.2. Above methods are applicable only to diagonal or triangular matrices and do not give the general answer to the question (2.1).

2.2. Certain Decomposition Methods for solving the system of linear equations

Here our aim is to write the coefficient matrix A as the product a lower triangular matrix L and an upper triangular matrix U and then go for solving $Ax = b$ or $LUx = b$, by solving two different systems. First $Lz = b$ for z and then $Ux = z$ for x .

2.3. Doolittle's Method

In this method A is decomposed as $A = LU$, where

$$L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}, \quad U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}.$$

So that

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = LU = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix} \quad (2.7)$$

Now one has to solve nine equations to find the all the total nine unknown coefficients of lower triangular matrix L and upper triangular matrix U . But these are easy to solve more or less only substitutions are needed. The solution is obtained by first solving $Lz = b$ for z by direct methods and then solving $Ux = z$ for x again by direct methods.

2.4. Crout's Method

Here one decomposes $A = LU$, where

$$L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}, \quad U = \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}.$$

Thus one has

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = LU = \begin{bmatrix} l_{11} & l_{11}u_{12} & l_{11}u_{13} \\ l_{21} & l_{21}u_{12} + l_{22} & l_{21}u_{13} + l_{22}u_{23} \\ l_{31} & l_{31}u_{12} + l_{32} & l_{31}u_{13} + l_{32}u_{23} + l_{33} \end{bmatrix} \quad (2.8)$$

Here we again need to solve nine equations to determine all the nine unknown coefficients. And similar to previous method we first solve $Lz = b$ for z and then $Ux = z$ for x .

2.5. Positive definite matrix

A real symmetric matrix A is said to be positive definite if $\det A > 0$ and all leading principal minors are positive.

2.6. A matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

is positive definite if $a_{21} = a_{12}$, $a_{23} = a_{32}$ and

- $a_{11} > 0$,
- $\det \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} > 0$.
- $\det A > 0$.

Example 2.1. The matrix $\begin{bmatrix} 1 & 1/2 & 1/3 \\ 1/2 & 1/3 & 1/4 \\ 1/3 & 1/4 & 1/5 \end{bmatrix}$ is positive definite, while the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 2 \\ 3 & 2 & 5 \end{bmatrix}$ is not positive definite because second leading principal minor is not positive.

Cholesky's Method

Cholesky's method is applicable for symmetric and positive definite matrix A . In this case the decomposition of A is $A = LL^T$, where

$$L = \begin{bmatrix} d_1 & 0 & 0 \\ l_{21} & d_2 & 0 \\ l_{31} & l_{32} & d_3 \end{bmatrix}.$$

So that

$$A = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{21} & a_{22} & a_{32} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = LU = \begin{bmatrix} d_1^2 & d_1 l_{12} & d_1 l_{13} \\ d_1 l_{21} & l_{21}^2 + d_2^2 & l_{21} l_{31} + d_2 l_{32} \\ d_1 l_{31} & l_{31} l_{21} + l_{32} d_2 & l_{31}^2 + l_{32}^2 + d_3^2 \end{bmatrix} \quad (2.9)$$

Here we only need to solve six equations in six unknowns. To solve $Ax = b$ we first solve $Lz = b$, for z and then $L^T x = z$ for x .

Question 2.2. Can we think of some generalization of the elimination method, which we learned in 10th standard?

2.7. Gauss elimination method

Here our aim is to convert the given system of linear equations $Ax = b$ in another equivalent (having the same solution) system of linear equations in which the coefficient matrix is of triangular or diagonal form. So that we can use the direct methods to solve it. Gauss elimination method involves three types of elementary row transformations, which does not change the solution of the system. These transformations can be described as follows.

1. Multiplication of a row by a non zero constant.

Suppose i_{th} row is multiplied by a non-zero constant c . We denote this transformation by $R_i(c)$. If we multiply the i_{th} row of the identity matrix by the same constant c and name this new matrix by $E_{i(c)}$, then it is easy to observe that the inverse of this matrix is $E_{i(c^{-1})}$. Now it is important to observe that this row transformation can also be operated to the system of linear equations by multiplying the system $Ax = b$ by elementary matrix $E_{i(c)}$ on the left to obtain $E_{i(c)}Ax = E_{i(c)}b$. And clearly this new system also have the solution as $x = A^{-1}(E_{i(c)})^{-1}E_{i(c)}b = A^{-1}b$.

2. Interchanging two rows.

The row transformation of interchanging the i_{th} row with the j_{th} row is denoted by R_{ij} . If we interchange the i_{th} row with the j_{th} row of the identity matrix and obtain a new matrix denoted by E_{ij} . This is a self inverse matrix. And the same row transformation can also be obtained by left multiplication of the elementary matrix E_{ij} on both sides of the system $Ax = b$.

3. Addition of non-zero multiple of one row to other.

Suppose we add a constant $c(\neq 0)$ multiple of i_{th} row to j_{th} row. This row transformation is denoted by $R_{ji(c)}$. The corresponding elementary matrix obtained from identity by the same row transformation is denoted by $E_{ji(c)}$. The inverse of this elementary matrix is $E_{ji(-c)}$.

Thus we note that at each step of elementary row transformation to a system of linear equations we get an equivalent system of linear equations having the same set of solution.

Steps of Gauss elimination method In case a priori we know that the coefficient matrix A is invertible, it can be converted to an upper triangular (or identity) matrix using the following steps.

- If $a_{11} = 0$, interchange the first row with j_{th} row for which $a_{j1} \neq 0$.
- Use transformations $R_{i1}(-a_{i1}/a_{11})$ to eliminate all a_{i1} , for all $i \neq 1$ with the pivot element a_{11} .
- If $a_{22} = 0$, interchange the second row with $j_{th}, j > 2$ row for which $a_{j2} \neq 0$
- Use transformations $R_{i2}(-a_{i2}/a_{22})$ to eliminate all a_{i2} for all $i \neq 2$ with the pivot element a_{22} .
- Use similar transformations to convert the coefficient matrix in to upper triangular matrix.

Also note that in this upper triangular matrix each diagonal entry is non-zero.

- Now by first using $R_{n(1/a_{nn})}$, we get $a_{nn} = 1$.
- Use transformations $R_{in}(-a_{in}), i = n-1, \dots, 1$ to eliminate $a_{in}, i = n-1, \dots, 1$ with pivot as $a_{nn} = 1$.
- We further do similar transformations in $(n-1)_{th}$ column after that in $(n-2)_{th}$ column and so on to second and then in first column. In general in k_{th} column we first apply $R_{k(1/a_{kk})}$ to get $a_{kk} = 1$ and then $R_{ik}(-a_{ik}), i = k-1, \dots, 1$ to eliminate $a_{ik}, i = k-1, \dots, 1$ with pivot as $a_{kk} = 1$.

Remark 2.3. Using Gauss elimination method we convert the matrix form of system of linear equations " $AX = b$ " by successive multiplying a finite number of elementary matrices on left to obtain an equivalent system

$$E_l E_{l-1} \cdots E_2 E_1 A X = E_l E_{l-1} \cdots E_2 E_1 b$$

such that $E_l E_{l-1} \cdots E_2 E_1 A = I$. Hence $A^{-1} = E_l E_{l-1} \cdots E_2 E_1$ and therefore converted system takes the form " $IX = A^{-1}b$ " to provide the solution to the system. In the process one can also obtain inverse of the matrix A as $E_l E_{l-1} \cdots E_2 E_1 I$ just by applying the same sequence of elementary row transformations to identity matrix.

Therefore, usually one start with augmented matrix $[A|b|I]$ and apply the same sequence of elementary row transformations to convert it to $[I|A^{-1}b|A^{-1}]$.

Observation 2.2. In general, the coefficient matrix might not be invertible or it is not a square matrix, that is, the number of equations is different from number of unknowns.

Question 2.3. Can we convert $Ax = b$ into some form, which can be solved directly when A is rectangular or singular matrix.

2.8. Row Echelon Form

Yes! One can still use elementary row transformations and Gauss elimination to convert the coefficient matrix A into Row reduced Echelon form.

A matrix A is said to be of Row reduced Echelon form if it satisfies the following properties.

- Each of the zero rows (a row in which each entry is zero), if it occurs in the matrix, must be placed below every non-zero row (a row in which at least one entry is non-zero).
- If the leading non-zero entry (first non-zero entry of the row) in i_{th} row occurs in k_{ith} column, that is, $a_{ij} = 0, j = 1, 2, \dots, k_i - 1$ and $a_{ik_i} \neq 0$, and there are only r many non-zero rows, then one must have $k_1 < k_2 < \dots < k_r$.

To obtain the row Echelon form of a matrix A of order $m \times n$, one can use the following steps.

- If first non-zero column is l_1 , by a row transformation (row interchange) bring one non-zero entry of l_{1th} column in the first row. And use this entry to eliminate all other entries of this column by the row transformations $R_{il_1}(-a_{il_1}/a_{1l_1})$. This l_1 is k_1 of the definition. In further row transformations we will not use the first row at all.
- Now consider the sub matrix of order $(m - 1) \times n$ after ignoring the first row. Search for the first non-zero column l_2 of this sub-matrix. If first entry of this l_{2th} column of order $m - 1 \times 1$ is zero, bring one non-zero entry of the column to the first row and use this entry to eliminate all other non-zero entries of the column. This l_2 is k_2 of the definition. In further row transformations we will not use the first two rows of the parent matrix.
- Now we further consider the sub matrix of order $(m - 2) \times n$ of the parent matrix. And search for first non-zero column l_3 of this sub-matrix. Use similar row transformations to eliminate all the entries of l_{3th} below the first row using the non-zero entry of the first row as pivot.
- Use similar transformations successively to find a column l_r of the parent matrix satisfying
 - i) l_{rth} column is the first non-zero column of the sub-matrix (obtained by ignoring first $r - 1$ rows) of order $(m - r + 1) \times n$,
 - ii) only non-zero entry of this column of sub-matrix is in the first row.
- This process will stop if i) either $r = m$, (in this case if $m = n$, we have row reduced Echelon form as upper triangular and invertible),
 - ii) or last $r - m$ rows of the parent matrix are zero rows.

2.9. Solution of system through Echelon form

For solving the system $Ax=b$, where A is coefficient matrix of order $m \times n$, x is column of unknowns of order $n \times 1$ and b is column of order $m \times 1$, we apply same sequence of elementary row transformations to augmented matrix $[A|b]$ to convert it in to $[A_E|b_E]$, where A_E is the Echelon form of matrix A . Because of elementary row transformations the solution x for both the systems $Ax = b$ and $A_E x = b_E$ are same.

We can also conclude about the nature of the solution by some observations of augmented matrix $[A_E|b_E]$.

- if there is a non-zero entry in the column matrix b_E in front of a zero row of A_E , then the system is inconsistent and there is no solution for the system.
- If $r = m$ and $n = m$, then unique solution by direct method.
- If $r = m$ and $n > m$, then infinite solutions with $n - m$ order of independency (degree of freedom), that is, out of n unknowns the system can be solved for any chosen m unknowns in terms of remaining $n - m$ unknowns, which can be given any values.

- Further if $r < m$ and last $m - r$ rows of augmented matrix $[A_E|b_E]$ are zero rows, then the system has unique solution if $n = r$, and infinite solutions with $n - r$ degree of freedom if $n > r$. Note that the case $n < r$ does not occur.

2.10. Pivoting

Example 2.2. Note that the exact solution for the following system of equations

$$0.0004x_1 + 1.402x_2 = 1.406$$

$$0.4003x_1 - 1.502x_2 = 2.501$$

is $x_1 = 10$ and $x_2 = 1$.

Note that if 0.0004 is used as pivot and computations are done with four significant places, then the first equation is multiplied by 1001 and subtracted from the second to obtain

$$-1001 \times 1.402 - 1.502)x_2 = 2.501 - 1001 \times 1.406$$

$$(-1403.402 - 1.502)x_2 = (2.501 - 1407.406)$$

$$\sim (-1403 - 1.502)x_2 = (2.501 - 1407)$$

$$-1405x_2 = -1404 \text{ gives } x_2 = 0.9993$$

if we use this value to get x_1 from first equation,

$$0.0004x_1 + 1.402 \times 0.9993 = 1.406$$

$$\sim 0.0004x_1 + 1.401 = 1.406$$

$$0.0004x_1 = 0.005, \text{ gives } x_1 = 12.5.$$

Thus the solution has the value of x_1 with significant error.

Observation 2.3. We carefully observe that this happens because in the process we are dividing by a number of very small magnitude, or equivalently multiplying by a big magnitude number to magnify the truncation error while storing.

Partial pivoting:

It is better to use pivot entry as of largest magnitude in the column. Thus if we use the largest magnitude entry of the column as pivot starting from the first column then second column and so on to last column. Such Gauss elimination method is known as partial pivoting.

Let us look at following system of equations

$$x_1 + x_2 + x_3 = 1$$

$$4x_1 + 3x_2 - x_3 = 6$$

$$3x_1 + 5x_2 + 3x_3 = 4$$

The augmented matrix $[A|b]$ for this system is

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 4 & 3 & -1 & 6 \\ 3 & 5 & 3 & 4 \end{array} \right]$$

Since 4 is the largest magnitude entry in the first column, we use this as pivot for the first column and operate $R_{12}(-\frac{1}{4})$ and $R_{32}(-\frac{3}{4})$ and get

$$\left[\begin{array}{ccc|c} 0 & 1/4 & 5/4 & -1/2 \\ 4 & 3 & -1 & 6 \\ 0 & 11/4 & 15/4 & -1/2 \end{array} \right]$$

Now for the reduced matrix $\left[\begin{array}{cc|c} 1/4 & 5/4 & -1/2 \\ 11/4 & 15/4 & -1/2 \end{array} \right]$, 11/4 is the largest magnitude entry in the first column. Therefore we operate $R_{13}(-1/11)$ in augmented matrix and get

$$\left[\begin{array}{ccc|c} 0 & 0 & 40/44 & -20/22 \\ 4 & 3 & -1 & 6 \\ 0 & 11/4 & 15/4 & -1/2 \end{array} \right]$$

Now first we operate R_{13} and then R_{12} to get upper triangular form of coefficient matrix.

$$\left[\begin{array}{ccc|c} 4 & 3 & -1 & 6 \\ 0 & 11/4 & 15/4 & -1/2 \\ 0 & 0 & 40/44 & -20/22 \end{array} \right]$$

And the solution can be obtained using backward substitution as

$x_3 = -1/2$, $x_2 = 1/2$ and $x_1 = 1$.

Complete pivoting

When one use the **largest magnitude entry of the matrix as pivot** to eliminate all other entries of that particular column having this pivot. and then **look for the next pivot as the largest entry of the reduced matrix** (obtained by crossing off the row and column having the first pivot). We apply the same process on further reduced matrix and so on.

Now if we look at the augmented matrix of previous example, we observe that 5 is the largest magnitude entry of the coefficient matrix, and hence we use this as our first pivot and operate $R_{13}(-1/5)$ and $R_{23}(-3/5)$ to get

$$\left[\begin{array}{ccc|c} 2/5 & 0 & 2/5 & 1/5 \\ 11/5 & 0 & -14/5 & 18/5 \\ 3 & 5 & 3 & 4 \end{array} \right]$$

Now for the reduced matrix $\left[\begin{array}{cc} 2/5 & 2/5 \\ 11/5 & -14/5 \end{array} \right]$, $-14/5$ is the largest magnitude entry, hence use this as next pivot and operate $R_{12}(1/7)$ to get

$$\left[\begin{array}{ccc|c} 25/35 & 0 & 0 & 25/35 \\ 11/5 & 0 & -14/5 & 18/5 \\ 3 & 5 & 3 & 4 \end{array} \right]$$

Note that in general, it is not possible to convert the coefficient matrix just by row transformations to either of the lower or upper triangular form (WRT main diagonal), but still one can keep track order of column-wise elimination and then use the reverse order of substitution with respect to unknowns.

In the above example, order of column-wise elimination is $2 \rightarrow 3 \rightarrow 1$, hence we should first evaluate x_1 from $1 - 1$ entry, then x_3 in terms of x_1 from $2 - 3$ entry and then finally, x_2 in terms of the previous two evaluated unknowns (x_1 and x_3) from $3 - 2$ entry of the matrix.

Remark 2.4. **One can interchange the columns to convert to lower triangular form.** Since column transformations can be done using right multiplication of elementary matrices, but to nullify this effect we need to left multiply by the inverse of the elementary matrix to the column of unknowns.

2.11. Solution of a homogeneous system of linear equations

Suppose all the terms appearing in the system are of degree one, that is, the constant appearing on the RHS of equations are zero and consequently the column vector b in the matrix form $Ax = b$ is a zero vector.

Recall that any $m \times n$ real matrix A represents a linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that

$$T(e_j) = \sum_{i=1}^m a_{ij} \bar{e}_i = \text{a vector represented by } j_{th} \text{ column of the matrix } A \text{ WRT SB of } \mathbb{R}^m,$$

where, $\{e_1, \dots, e_j, \dots, e_n\}$ is a standard basis of $\mathbb{R}^n$ and $\{\bar{e}_1, \dots, \bar{e}_i, \dots, \bar{e}_m\}$ is standard basis for $\mathbb{R}^m$.

Now, finding solutions to $Ax = 0$, is equivalent to finding the collection of vectors ' x ' satisfying $T(x) = 0$, that is the null space of T .

Observation 2.4. It is clear that if two vectors X_1 and X_2 satisfy, $AX = 0$, we have $AX_1 = 0, AX_2 = 0$ and hence $A(\alpha X_1 + \beta X_2) = 0$, shows that the collection of solutions is a vector space.

In particular, if $A = \begin{bmatrix} 1 & -2 & -1 \\ 2 & -4 & -2 \\ -3 & 6 & 3 \end{bmatrix}$, two independent solutions to $AX = 0$ are $X_1 = [2, 1, 0]^t$ and $X_2 = [1, 0, 1]^t$ and the plane generated by these vectors is $x - 2y - z = 0$, that represents the solution vector space.

Question 2.4. Whether the solution to $AX = 0$ and $AX = b$ are related to each other.

Theorem 2.2. Suppose S is the solution space to $AX = 0$ and X_0 is a particular solution to $AX = b$, then $S + X_0 = \{X + X_0 : X \in S\}$ is the collection of all possible solutions to $AX = b$.

Proof. Note that if $X \in S$, we have $AX = 0$ and $A(X + X_0) = AX + AX_0 = 0 + b = b$, hence each element of $S + X_0$ is the solution to $AX = b$.

Next if Y is a solution to $AX = b$, that is, $AY = b$, we must have $A(Y - X_0) = AY - AX_0 = b - b = 0$, hence $Y - X_0$ is in S , or, $Y \in S + X_0$. $\square$

Remark 2.5. Geometrically, if S is the solution plane (passing through origin) for $AX = 0$, then the solution of $AX = b$ is a plane parallel to S and passing through X_0 .

Keeping A as in the previous example, we look for the solution to the system $AX = b$ where $b = [3, 6, -9]^t$. One particular solution to $AX = b$ is $[1, -1, 0]^t$, hence its complete solution is the plane $x - 2y - z = 3$.

2.12. Norm

Note that real line (being a vector space) has a notion of length of a vector, denoted by mode $|x| = \sqrt{x^2}$, e.g., $|3| = 3$ and $|-5| = 5$.

In $\mathbb{R}^2$, the Euclidean length of a vector $X = (x_1, x_2)$ is the distance of the point X from the origin $O = (0, 0)$, denoted by $\|X\| = \sqrt{x_1^2 + x_2^2}$.

Exercise 2.1. Show that the norm $\|X\| = \sqrt{x_1^2 + x_2^2}$ for all $X = (x_1, x_2) \in \mathbb{R}^2$ satisfies the following properties

- $\|X\| \geq 0$ and $\|X\| = 0$ if and only if $X = O$, for all $X \in \mathbb{R}^2$,
- $\|\alpha \odot X\| = |\alpha| \|X\|$, for any scalar $\alpha \in \mathbb{R}$ and $X \in \mathbb{R}^2$, where $\alpha \cdot X = (\alpha x_1, \alpha x_2) \in V$,
- $\|X \oplus Y\| \leq \|X\| + \|Y\|$ for all $X, Y \in \mathbb{R}^2$, where $X \oplus Y = (x_1 + y_1, x_2 + y_2) \in V$.

Definition:

Let $V(\oplus, \odot)$ be a vector space over the field of real numbers $\mathbb{R}$. A function

$\|x\| : V \rightarrow [0, \infty]$ for all $x \in V$

satisfying the following properties

- a) $\|x\| \geq 0$ and $\|x\| = 0$ if and only if $x = 0$, for all $x \in V$,
- b) $\|\alpha \odot x\| = |\alpha| \|x\|$, for any scalar $\alpha \in \mathbb{R}$ and $x \in V$,
- c) $\|x \oplus y\| \leq \|x\| + \|y\|$ for all $x, y \in V$

is called the **norm** of the vector $x \in V$, and the linear space $V(\oplus, \odot, \|\cdot\|)$ is called the **Normed Linear Space**.

Example 2.3. Consider the function $\|\cdot\|_p$ on the vector space $\mathbb{R}^n$ to $[0, \infty)$ defined by

$$\|X\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}, \text{ for all } X = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n. \quad (2.10)$$

This defines a norm on $\mathbb{R}^n$, known as **p-norm of the vector $X \in \mathbb{R}^n$** . Note that in daily life we use 2-norm for measuring the distance in plane and space.

Example 2.4. Consider the function on $\mathbb{R}^n$

$$\|X\|_\infty = \max_{1 \leq i \leq n} \{|x_i|\}. \quad (2.11)$$

This also defines a norm on $\mathbb{R}^n$.

Exercise 2.2. Draw the locus of following in a two dimensional plane

- $\{x \in \mathbb{R}^2 : \|x\|_1 = 1\}$,
- $\{x \in \mathbb{R}^2 : \|x\|_2 = 1\}$,
- $\{x \in \mathbb{R}^2 : \|x\|_\infty = 1\}$.

Observation 2.5. Note that the euclidean distance between any two vectors X and Y in $\mathbb{R}^2$ can be obtained as the length of the vector $X - Y$, defined by $\|X - Y\| = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$ and satisfy the following properties

- a) $\|X - Y\| = 0 \Leftrightarrow X = Y$, for all $X, Y \in \mathbb{R}^2$,
- b) $\|X - Y\| = \|Y - X\|$, for all $X, Y \in \mathbb{R}^2$,
- c) $\|X - Y\| \leq \|X - Z\| + \|Z - Y\|$, for all $X, Y, Z \in \mathbb{R}^2$.

Definition: Distance (/Metric)

Let M be a nonempty set. A function $d : M \times M \rightarrow [0, \infty)$ is called the distance (/metric) on the set M if it satisfies the following properties

- a) $d(x, y) = 0 \Leftrightarrow x = y$ for all $x, y \in M$,
- b) $d(x, y) = d(y, x)$ for all $x, y \in M$,
- c) Triangle Inequality: $d(x, y) \leq d(x, z) + d(z, y)$ for all $x, y, z \in M$.

The space (M, d) is called the metric space equipped with the distance d .

Definition:

A function f from a metric space (M_1, d_1) to (M_2, d_2) is called **continuous** at a point $x \in M_1$ if for $\epsilon > 0$ there exists a $\delta > 0$ such that $y \in M_1$ and $d_1(x, y) < \delta$ implies that $d_2(f(x), f(y)) < \epsilon$.

A function f from a metric space (M_1, d_1) to (M_2, d_2) is called **pointwise continuous** if f is continuous for all $x \in M_1$.

A function f from a metric space (M_1, d_1) to (M_2, d_2) is called **uniformly continuous** if for $\epsilon > 0$ there exists a $\delta > 0$ such that $x, y \in M_1$ and $d_1(x, y) < \delta$ implies that $d_2(f(x), f(y)) < \epsilon$.

A function f from a metric space (M_1, d_1) to (M_2, d_2) is called **Lipschitz continuous** if there is a $K > 0$ such that $d_2(f(x), f(y)) < K d_1(x, y)$ for all $x, y \in M_1$.

Exercise 2.3. Show that **Lipschitz continuity** $\Rightarrow$ **uniformly continuity** $\Rightarrow$ **pointwise continuity**.

Exercise 2.4. Let $V(\oplus, \odot, \|\cdot\|)$ be a Normed Linear Space. Show that the function $d : V \times V \rightarrow [0, \infty)$, defined by $d(x, y) = \|x - y\|$ is a distance (induced by the norm) on V .

Observation 2.6. Using this exercise, one can define a distance function d_p on $\mathbb{R}^n$ as $d_p(x, y) = \|x - y\|_p$ for all $1 \leq p \leq \infty$.

Exercise 2.5. Show that a sequence $\{x_k\}$ in $(\mathbb{R}^n, \|\cdot\|_p)$ converges to $x \in \mathbb{R}^n$ if and only if $(x_k)_i \rightarrow x_i$ for all $1 \leq i \leq n$.

Exercise 2.6. Show that norm $\|\cdot\|_p$ is a continuous function from $\mathbb{R}^n \rightarrow \mathbb{R}$ for all $1 \leq p \leq \infty$.

Recall that any matrix A of order $m \times n$ represents a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$, and hence A can also serve as a map from $\mathbb{R}^n$ to $\mathbb{R}^m$ and we can ask about the continuity of A from $\mathbb{R}^n$ to $\mathbb{R}^m$, when they are equipped with specified distances.

Definition:

Suppose A is a real matrix of order $m \times n$ and representing a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$, and suppose both spaces are equipped with $\|\cdot\|_p$. Define

$$\|A\|_p = \sup\{\|Ay\|_p : \|y\|_p \leq 1\}$$

Fact: $\|A\|_p$ defines a matrix norm on M_{mn} .

Problem 2.1. Suppose A is a real matrix of order $m \times n$. Show that $\|Ax\|_p \leq \|A\|_p \|x\|_p$ for all $x \in \mathbb{R}^n$

Solution. Note that if $x \in \mathbb{R}^n$, and $y = \frac{x}{\|x\|_p}$, then $\|y\|_p = 1$, hence

$$\|A\|_p \geq \|Ay\|_p = \|A(\frac{x}{\|x\|_p})\|_p = \frac{1}{\|x\|_p} \|Ax\|_p. \text{ This implies that}$$

$$\|Ax\|_p \leq \|A\|_p \|x\|_p.$$

Exercise 2.7. Show that for any real matrix of order $m \times n$,

$$\|A\|_1 = \max_{1 \leq j \leq n} \left\{ \left(\sum_{i=1}^m |a_{ij}| \right) : j = 1, \dots, n \right\}, \text{ and} \quad (2.12)$$

$$\|A\|_\infty = \max_{1 \leq i \leq m} \left\{ \left(\sum_{j=1}^n |a_{ij}| \right) : i = 1, \dots, m \right\}. \quad (2.13)$$

Problem 2.2. Let A be a real matrix of order $m \times n$ and B be a real matrix of order $n \times k$. Show that

$$\|BA\|_p \leq \|B\|_p \|A\|_p.$$

Solution. Note that A is a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ and B is a linear transformation from $\mathbb{R}^m$ to $\mathbb{R}^k$ and the matrix BA is the composite linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^k$

$$(\mathbb{R}^n, \|\cdot\|_p) \xrightarrow{A} (\mathbb{R}^m, \|\cdot\|_p) \xrightarrow{B} (\mathbb{R}^k, \|\cdot\|_p).$$

For any vector $x \in \mathbb{R}^n$, we have $BAx \in \mathbb{R}^k$ and

$$\|BAx\|_p \leq \|B\|_p \|Ax\|_p \leq \|B\|_p \|A\|_p \|x\|_p.$$

Thus for any vector $x \in \mathbb{R}^n$ with $\|x\|_p \leq 1$, we have $\|BAx\|_p \leq \|B\|_p \|A\|_p \|x\|_p \leq \|B\|_p \|A\|_p$ and the set $\{\|BAx\|_p : \|x\|_p \leq 1\}$ is bounded by $\|B\|_p \|A\|_p$. Hence $\|BA\|_p \leq \|A\|_p \|B\|_p$.

Problem 2.3. If λ is an eigenvalue of the matrix A of order $n \times n$, then show that

$$|\lambda| \leq \|A\|_p.$$

Solution. Let x be the eigenvector corresponding to the eigenvalue λ . Hence $Ax = \lambda x$ and $|\lambda| \|x\|_p = \|\lambda x\|_p = \|Ax\|_p \leq \|A\|_p \|x\|_p$. Since x is a nonzero vector $\|x\|_p \neq 0$ and we have $|\lambda| \leq \|A\|_p$.

Definition:

The **spectral radius** of a square matrix A is denoted by $\rho(A)$ and defined as

$$\rho(A) = \max\{|\lambda| : \lambda \text{ is an eigenvalue of the matrix } A\}.$$

Exercise 2.8. Let A be a square matrix. Show that $\rho(A) < \|A\|_p$ for any $1 \leq p \leq \infty$.

Definition: **Strict Row Diagonally Dominant matrix SRDD** :
 Let $A = [a_{ij}]$ be of order $n \times n$. We say that A is SRDD if $|a_{ii}| > \sum_{j=1, j \neq i}^n |a_{ij}|$ for all $1 \leq i \leq n$.

Exercise 2.9. Show that if A is SRDD matrix, then $|a_{ii}| > 0$ for all $1 \leq i \leq n$ and A is invertible.

2.13. Gauss-Jacobi iterative method

Now our aim is to solve (2.1) using some iteration method. For this we first assume that A is SRDD so that $a_{ii} \neq 0$ and rewrite the system of linear equations as follows

$$\begin{array}{rcll}
 a_{11}x_1 & = & 0x_1 & -a_{12}x_2 \quad \dots \quad \dots \quad \dots \quad -a_{1i}x_i \quad \dots \quad \dots \quad \dots \quad -a_{1n}x_n & +b_1 \\
 a_{22}x_2 & = & -a_{21}x_1 & 0x_2 \quad \dots \quad \dots \quad \dots \quad -a_{2i}x_i \quad \dots \quad \dots \quad \dots \quad -a_{2n}x_n & +b_2 \\
 \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\
 \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\
 a_{ii}x_i & = & -a_{i1}x_1 & -a_{i2}x_2 \quad \dots \quad \dots \quad \dots & 0x_i & \dots \quad \dots \quad \dots & -a_{in}x_n & +b_i \\
 \dots & = & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\
 \dots & = & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\
 \dots & = & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\
 a_{nn}x_n & = & -a_{n1}x_1 & -a_{n2}x_2 \quad \dots \quad \dots \quad \dots & -a_{ni}x_i & \dots \quad \dots \quad \dots & 0x_n & +b_n
 \end{array}$$

Note that if we decompose $A = L + D + U$, where L, D, U are lower triangular, diagonal and upper triangular matrices respectively, then above system of equations in matrix form can be written as $Dx = -(L + U)x + b$, where x is the column vector of unknowns. Now since D is invertible by our assumption we have

$$x = -D^{-1}(L + U)x + D^{-1}b. \quad (2.14)$$

And this can also be written component wise as

$$\begin{array}{rcll}
 x_1 & = & 0x_1 & -\frac{a_{12}}{a_{11}}x_2 \quad \dots \quad \dots \quad \dots \quad -\frac{a_{1i}}{a_{11}}x_i \quad \dots \quad \dots \quad \dots \quad -\frac{a_{1n}}{a_{11}}x_n & +\frac{b_1}{a_{11}} \\
 x_2 & = & -\frac{a_{21}}{a_{22}}x_1 & 0x_2 \quad \dots \quad \dots \quad \dots \quad -\frac{a_{2i}}{a_{22}}x_i \quad \dots \quad \dots \quad \dots \quad -\frac{a_{2n}}{a_{22}}x_n & +\frac{b_2}{a_{22}} \\
 \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\
 \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\
 \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\
 x_i & = & -\frac{a_{i1}}{a_{ii}}x_1 & -\frac{a_{i2}}{a_{ii}}x_2 \quad \dots \quad \dots \quad \dots & 0x_i & \dots \quad \dots \quad \dots & -\frac{a_{in}}{a_{ii}}x_n & +\frac{b_i}{a_{ii}} \\
 \dots & = & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\
 \dots & = & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\
 \dots & = & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots \\
 x_n & = & -\frac{a_{n1}}{a_{nn}}x_1 & -\frac{a_{n2}}{a_{nn}}x_2 \quad \dots \quad \dots \quad \dots & -\frac{a_{ni}}{a_{nn}}x_i & \dots \quad \dots \quad \dots & 0x_n & +\frac{b_n}{a_{nn}}
 \end{array} \quad (2.15)$$

Denoting $-D^{-1}(L + U) = H$ and $D^{-1}b = C$, we have $x = Hx + C$. Now if we assume the initial approximation to the solution as

$$x^{(0)} = \begin{bmatrix} x_1^{(0)} & x_2^{(0)} & \dots & \dots & \dots & x_i^{(0)} & \dots & \dots & \dots & x_n^{(0)} \end{bmatrix}^t, \quad (2.16)$$

then the first approximation can be obtained by the equation $x^{(1)} = Hx^{(0)} + C$, and in general $(k + 1)_{th}$ approximation is obtained by

$$x^{(k+1)} = Hx^{(k)} + C, \quad (2.17)$$

where matrix H and C are as follows

$$H = \begin{bmatrix} 0 & -\frac{a_{12}}{a_{11}} & \cdots & \cdots & \cdots & -\frac{a_{1i}}{a_{11}} & \cdots & \cdots & \cdots & -\frac{a_{1n}}{a_{11}} \\ -\frac{a_{21}}{a_{22}} & 0 & \cdots & \cdots & \cdots & -\frac{a_{2i}}{a_{22}} & \cdots & \cdots & \cdots & -\frac{a_{2n}}{a_{22}} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ -\frac{a_{i1}}{a_{ii}} & -\frac{a_{i2}}{a_{ii}} & \cdots & \cdots & \cdots & 0 & \cdots & \cdots & \cdots & -\frac{a_{in}}{a_{ii}} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ -\frac{a_{n1}}{a_{nn}} & -\frac{a_{n2}}{a_{nn}} & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & 0 \end{bmatrix}, \quad C = \begin{bmatrix} -\frac{a_{1n}}{a_{11}} \\ -\frac{a_{2n}}{a_{22}} \\ \cdots \\ \cdots \\ \cdots \\ -\frac{a_{in}}{a_{ii}} \\ \cdots \\ \cdots \\ \cdots \\ 0 \end{bmatrix}. \quad (2.18)$$

Using (2.17) we can write i_{th} component of the $(k+1)_{th}$ approximation of the solution vector as

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left[b_i - \sum_{j=1, j \neq i}^n a_{ij} x_j^{(k)} \right]. \quad (2.19)$$

2.14. Gauss-Seidel iterative method

We rewrite the system of equations as follows

$$\begin{array}{cccccccccccccccc} a_{11}x_1 & = & -a_{12}x_2 & -a_{13}x_3 & \cdots & \cdots & -a_{1i}x_i & \cdots & \cdots & \cdots & \cdots & -a_{1n}x_n & +b_1 \\ a_{21}x_1 & +a_{22}x_2 & = & -a_{23}x_3 & \cdots & \cdots & -a_{2i}x_i & \cdots & \cdots & \cdots & \cdots & -a_{2n}x_n & +b_2 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ a_{i1}x_1 & +a_{i2}x_2 & \cdots & \cdots & \cdots & +a_{ii}x_i & = & -a_{i,i+1}x_{i+1} & \cdots & \cdots & \cdots & -a_{in}x_n & +b_i \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ a_{n1}x_1 & +a_{n2}x_2 & \cdots & \cdots & \cdots & +a_{ni}x_i & \cdots & \cdots & \cdots & \cdots & +a_{nn}x_n & = & +b_n \end{array}$$

For Gauss-Seidel method we again assume that A is SRDD so that $a_{ii} \neq 0$ and we decompose $A = L + D + U$, then above system of equations in matrix form can be written as $(L + D)x = -Ux + b$, where the matrix $L + D$ turns out to be invertible. Further if $x^{(0)}$ is the initial approximation, we define the sequence of iterates by

$$(L + D)x^{(k+1)} = -Ux^{(k)} + b, \quad \text{or} \quad (2.20)$$

$$x^{(k+1)} = -(L + D)^{-1}Ux^{(k)} + (L + D)^{-1}b. \quad (2.21)$$

Thus the Gauss-Seidel iteration scheme can be written as

$$x^{(k+1)} = Hx^{(k)} + C, \quad (2.22)$$

where $H = -(L + D)^{-1}U$ and $C = (L + D)^{-1}b$.

Component-wise (2.20) can be written as follows

$$\sum_{j=1}^{i-1} a_{ij}x_j^{(k+1)} + a_{ii}x_i^{(k+1)} = - \sum_{j=i+1}^n a_{ij}x_j^{(k)} + b_i, \quad \text{or equivalently}$$

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left[b_i - \sum_{j=1}^{i-1} a_{ij}x_j^{(k+1)} - \sum_{j=i+1}^n a_{ij}x_j^{(k)} \right]. \quad (2.23)$$

2.15. Convergence of Iteration Scheme $x^{(k+1)} = Hx^{(k)} + C$

Note that both Gauss-Jacobi and Gauss-Seidel iteration schemes are of the form (2.17). The convergence of such scheme is necessary to approximate the solution vector.

Theorem 2.3. *The iteration scheme $x^{(k+1)} = Hx^{(k)} + C$ converges to $(I - H)^{-1}C$ if and only if $\rho(H) < 1$, where $\rho(H)$ is the spectral radius of the matrix H .*

Proof. (outline of the proof) We would present the outline of the proof for the simple case when H is diagonalisable such that there exists an orthogonal matrix S such that $S^{-1}HS = D$, where D is the diagonal matrix, consisting of its n diagonal entries as the eigenvalues of H repeated with their multiplicity.

Note that if $x^{(0)}$ is the initial approximation to solution vector x , we have first approximation $x^{(1)} = Hx^{(0)} + C$ and second as $x^{(2)} = Hx^{(1)} + C = H(Hx^{(0)} + C) + C = H^2x^{(0)} + [I + H]C$, and in general we have,
 $x^{(k)} = H^kx^{(0)} + [I + H + H^2 + \dots + H^{k-1}]C$.

Therefore, convergence of the sequence $\{x^{(k)}\}$ is equivalent to simultaneous convergence of the sequence $\{H^k\}$ and the convergence of the series $I + H + H^2 + \dots + H^{k-1} + \dots$.

Note that $(I + H + H^2 + \dots + H^{k-1})(I - H) = I - H^k$, therefore convergence of the series is equivalent to convergence of H^k .

Further since, $H^k = (SDS^{-1})^k = SD^kS^{-1}$, convergence of H^k is equivalent to convergence of D^k and that is equivalent to convergence of λ_i^k for all i , and that can be guaranteed by the assumption $|\lambda_i| < 1$ for all i . This can be ensured by the assumption $\rho(H) < 1$. [For operational purpose we will assume that $\|H\|_\infty < 1$, because $\rho(H) \leq \|H\|_\infty$.] $\square$

Theorem 2.4. *Gauss-Jacobi iterates converges to $A^{-1}b$.*

Proof. We have $H = -D^{-1}(L + U)$ and $C = D^{-1}b$, where $A = L + D + U$. Therefore,

$$H = \begin{bmatrix} 0 & -\frac{a_{12}}{a_{11}} & \dots & -\frac{a_{1n}}{a_{11}} \\ -\frac{a_{21}}{a_{22}} & 0 & \dots & -\frac{a_{2n}}{a_{22}} \\ \vdots & \vdots & \ddots & \vdots \\ -\frac{a_{n1}}{a_{nn}} & -\frac{a_{n2}}{a_{nn}} & \dots & 0 \end{bmatrix}.$$

Since A is SRDD, we have $\sum_{j=1, j \neq i}^n |a_{ij}| < |a_{ii}|$ for all $1 \leq i \leq n$, or $\sum_{j=1, j \neq i}^n \frac{|a_{ij}|}{|a_{ii}|} < 1$, and therefore $\|H\|_\infty < 1$, because $\|H\|_\infty = \max_{1 \leq i \leq n} \sum_{j=1, j \neq i}^n \frac{|a_{ij}|}{|a_{ii}|}$. Therefore, $\rho(H) \leq \|H\|_\infty < 1$ implies that the sequence of iterates in Gauss-Jacobi method converges to $(I - H)^{-1}C$. Thus it is enough to prove that $(I - H)^{-1}C = A^{-1}b$. Note that
 $(I - H)^{-1}C = [I + D^{-1}(L + U)]^{-1}(D^{-1}b) = [D^{-1}(D + L + U)]^{-1}(D^{-1}b) = A^{-1}DD^{-1}b = A^{-1}b$. $\square$

Theorem 2.5. *Gauss-Seidel iterates converges to $A^{-1}b$.*

Proof. We have $H = -(L + D)^{-1}U$ and $C = (L + D)^{-1}b$, where $A = L + D + U$.

Claim: $\|H\|_\infty < 1$.

Let $x \in \mathbb{R}^n$ such that $\|x\|_\infty \leq 1$. Let $y = Hx$. Therefore, $y = -(L + D)^{-1}Ux$, or, $(L + D)y = -Ux$, or $y = -D^{-1}Ly - D^{-1}Ux$. Therefore,

$$\begin{bmatrix} y_1 \\ \vdots \\ y_{i-1} \\ y_i \\ \vdots \\ y_n \end{bmatrix} = - \begin{bmatrix} 0 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & 0 \\ \frac{a_{i-1,1}}{a_{i-1,i-1}} & \dots & 0 & 0 & \dots & 0 \\ \frac{a_{i1}}{a_{ii}} & \dots & \frac{a_{i,i-1}}{a_{ii}} & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & 0 \\ \frac{a_{n1}}{a_{nn}} & \dots & \frac{a_{n,i-1}}{a_{nn}} & \frac{a_{ni}}{a_{nn}} & \dots & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_{i-1} \\ y_i \\ \vdots \\ y_n \end{bmatrix} - \begin{bmatrix} 0 & \dots & \frac{a_{1,i-1}}{a_{11}} & \frac{a_{1i}}{a_{11}} & \dots & \frac{a_{1n}}{a_{11}} \\ \vdots & \ddots & \vdots & \vdots & \ddots & 0 \\ 0 & \dots & 0 & \frac{a_{i-1,i}}{a_{i-1,i-1}} & \dots & \frac{a_{i-1,n}}{a_{i-1,i-1}} \\ 0 & \dots & 0 & 0 & \dots & \frac{a_{in}}{a_{ii}} \\ \vdots & \ddots & \vdots & \vdots & \ddots & 0 \\ 0 & \dots & 0 & 0 & \dots & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_{i-1} \\ x_i \\ \vdots \\ x_n \end{bmatrix}.$$

Comparing the i_{th} component, we have

$$y_i = -\sum_{j=1}^{i-1} \frac{a_{ij}}{a_{ii}} y_j - \sum_{j=i+1}^n \frac{a_{ij}}{a_{ii}} x_j, \text{ for all } 1 \leq i \leq n,$$

$$\Rightarrow |y_i| \leq \sum_{j=1}^{i-1} \frac{|a_{ij}|}{|a_{ii}|} |y_j| + \sum_{j=i+1}^n \frac{|a_{ij}|}{|a_{ii}|} |x_j| \leq \sum_{j=1}^{i-1} \frac{|a_{ij}|}{|a_{ii}|} \|y\|_\infty + \sum_{j=i+1}^n \frac{|a_{ij}|}{|a_{ii}|} \|x\|_\infty,$$

Therefore, for each i we have,

$$|y_i| \leq \sum_{j=1}^{i-1} \frac{|a_{ij}|}{|a_{ii}|} \|y\|_\infty + \sum_{j=i+1}^n \frac{|a_{ij}|}{|a_{ii}|}$$

Suppose if i_0 is such that $\|y\|_\infty = |y_{i_0}|$, then we have (i_0 depends on x)

$$\|y\|_\infty = |y_{i_0}| \leq \sum_{j=1}^{i_0-1} \frac{|a_{i_0 j}|}{|a_{i_0 i_0}|} \|y\|_\infty + \sum_{j=i_0+1}^n \frac{|a_{i_0 j}|}{|a_{i_0 i_0}|}, \text{ therefore}$$

$$\|y\|_\infty \leq \frac{\sum_{j=i_0+1}^n \frac{|a_{i_0 j}|}{|a_{i_0 i_0}|}}{1 - \sum_{j=1}^{i_0-1} \frac{|a_{i_0 j}|}{|a_{i_0 i_0}|}} = \frac{\beta_{i_0}}{1 - \alpha_{i_0}}, \text{ where}$$

$$\alpha_i = \sum_{j=i+1}^n \frac{|a_{ij}|}{|a_{ii}|}, \alpha_n = 0 \text{ and } \beta_i = \sum_{j=1}^{i-1} \frac{|a_{ij}|}{|a_{ii}|}, \beta_1 = 0 \text{ for all } 1 \leq i \leq n.$$

Define $\eta = \max_{1 \leq i \leq n} \frac{\beta_i}{1 - \alpha_i}$. Then we have

$$\|y\|_\infty = \|Hx\|_\infty \leq \eta$$

Note that η does not depend on the chosen x , hence we have

$$\|y\|_\infty = \|Hx\|_\infty \leq \eta \text{ for all } \|x\|_\infty \leq 1, x \in \mathbb{R}^n. \text{ Hence } \rho(H) \leq \|H\|_\infty \leq \eta.$$

Thus to conclude the convergence of iterates, it is sufficient to show that $\eta < 1$.

Claim: $\eta < 1$.

Since A is SRDD, we have $\sum_{j=1, j \neq i}^n |a_{ij}| < |a_{ii}|$ for all $1 \leq i \leq n$, and hence for each i we have or $\sum_{j=1}^{i-1} \frac{|a_{ij}|}{|a_{ii}|} + \sum_{j=i+1}^n \frac{|a_{ij}|}{|a_{ii}|} < 1$, or $\alpha_i + \beta_i < 1$, or $\frac{\beta_i}{1 - \alpha_i} < 1$. Therefore we have $\eta < 1$ and the sequence of iterates in Gauss-Seidel method must converge to $(I - H)^{-1}C$. Thus it is enough to prove that $(I - H)^{-1}C = A^{-1}b$. Note that $(I - H)^{-1}C = [I + (L + D)^{-1}U]^{-1}(L + D)^{-1}b = [(L + D)^{-1}(L + D + U)]^{-1}(L + D)^{-1}b = A^{-1}b$. $\square$

2.16. Error analysis of Iteration Methods

Suppose x is the exact solution to the system (2.17). Then the k_{th} error vector can be defined as

$$e^{(k)} = x - x^{(k)} = \begin{bmatrix} x_1 - x_1^{(k)} & x_2 - x_2^{(k)} & \dots & \dots & x_i - x_i^{(k)} & \dots & \dots & x_n - x_n^{(k)} \end{bmatrix}^t, \quad (2.24)$$

where $[a_{ij}]^t$ represents the transpose of the corresponding matrix. Using (2.24) and (2.19), one has

$$e^{(k+1)} = x - x^{(k+1)} = (Hx + C) - (Hx^{(k)} + C) = H(x - x^{(k)}) = He^{(k)}. \quad (2.25)$$

Therefore, $\|e^{(k+1)}\|_\infty = \|He^{(k)}\|_\infty \leq \|H\|_\infty \|e^{(k)}\|_\infty$ for all $k \in \mathbb{N}$. And repeatedly using this fact we have

$$\|e^{(k+1)}\|_\infty \leq \|H\|_\infty^k \|e^{(1)}\|_\infty. \quad (2.26)$$

Therefore, for each $k \in \mathbb{N}$, we have

$$\|e^{(k)}\|_\infty \leq \mu^k \|e^{(0)}\|_\infty \quad \text{for Gauss-Jacobi Method} \quad (2.27)$$

$$\|e^{(k)}\|_\infty \leq \eta^k \|e^{(0)}\|_\infty \quad \text{for Gauss-Seidel Method} \quad (2.28)$$

Thus, we have $\|e^{(k)}\|_\infty \rightarrow 0$.

Remark 2.6. Note that $x^{(k+1)} - x^{(k)} = Hx^{(k)} + C - Hx^{(k-1)} - C$ and hence for all $k \in \mathbb{N}$,

$$\|x^{(k+1)} - x^{(k)}\|_\infty = \|H(x^{(k)} - x^{(k-1)})\|_\infty \leq \|H\|_\infty \times \|x^{(k)} - x^{(k-1)}\|_\infty.$$

Thus $\|x^{(k+1)} - x^{(k)}\|_\infty \leq \|H\|_\infty \times \|H\|_\infty \|x^{(k-1)} - x^{(k-2)}\|_\infty \leq \dots \leq \|H\|_\infty^k \|x^{(1)} - x^{(0)}\|_\infty$.

Now for any $m > k > 1$ we have

$$\begin{aligned} \|x^{(m)} - x^{(k)}\|_\infty &= \|x^{(m)} - x^{(m-1)} + x^{(m-1)} - x^{(m-2)} + x^{(m-2)} - \dots - x^{(k+1)} + x^{(k+1)} - x^{(k)}\|_\infty \\ &\leq \|x^{(m)} - x^{(m-1)}\|_\infty + \|x^{(m-1)} - x^{(m-2)}\|_\infty + \dots + \|x^{(k+1)} - x^{(k)}\|_\infty \\ &\leq (\|H\|_\infty^{m-1} + \|H\|_\infty^{m-2} + \dots + \|H\|_\infty^k) \|x^{(1)} - x^{(0)}\|_\infty \\ &< \|H\|_\infty^k \left(\sum_{i=0}^{\infty} \|H\|_\infty^i \right) \|x^{(1)} - x^{(0)}\|_\infty \leq \frac{\|H\|_\infty^k}{1 - \|H\|_\infty} \|x^{(1)} - x^{(0)}\|_\infty. \end{aligned}$$

Further since $\|x^{(m)} - x\|_\infty \rightarrow 0$, for any given ϵ , there exists an integer $m_0 > k$ such that $\|x^{(m_0)} - x\|_\infty < \epsilon$, therefore

$$\|x - x^{(k)}\|_\infty \leq \|x - x^{(m_0)}\|_\infty + \|x^{(m_0)} - x^{(k)}\|_\infty < \epsilon + \frac{\|H\|_\infty^k}{1 - \|H\|_\infty} \|x^{(1)} - x^{(0)}\|_\infty.$$

Since above inequality is true for any arbitrary $\epsilon > 0$, we can assume $\epsilon \rightarrow 0$ and hence

$$\|e^{(k)}\|_\infty = \|x - x^{(k)}\|_\infty \leq \frac{\|H\|_\infty^k}{1 - \|H\|_\infty} \|x^{(1)} - x^{(0)}\|_\infty. \quad (2.29)$$

Therefore, we have

$$\|e^{(k)}\|_\infty \leq \frac{\mu^k}{1-\mu} \|x^{(1)} - x^{(0)}\|_\infty \text{ for Gauss-Jacobi Method and}$$

$$\|e^{(k)}\|_\infty \leq \frac{\eta^k}{1-\eta} \|x^{(1)} - x^{(0)}\|_\infty \text{ for Gauss-Jacobi Method because } \|H\|_\infty \leq \eta.$$

Exercise 2.10. If $\mu < 1$, prove that $\eta \leq \mu$.

Remark 2.7. Above exercise shows that if the coefficient matrix is strict row diagonally dominant, then the convergence factor of Gauss-Jacobi method is less or equal to convergence factor of Gauss-Seidel method. And in case if $\eta < \mu$, Gauss-Seidel method should converges faster than Gauss-Jacobi method.

Theorem 2.6. Suppose A and B are two square matrices of order $n \times n$. If A is invertible and

$$\|A - B\| < \frac{1}{\|A^{-1}\|},$$

then B is also invertible.

Proof. Suppose if B is not invertible, then the rank of B is less than n and hence by Rank-Nullity theorem the null space of B is not equal to $\{0\}$. Thus if $0 \neq x$ is in the Null space of (B) , then $Bx = 0$ and

$$\frac{\|x\|}{\|A^{-1}\|} = \frac{\|A^{-1}Ax\|}{\|A^{-1}\|} \leq \|Ax\| = \|Ax - Bx\| \leq \|A - B\| \times \|x\|.$$

But $\|x\| \neq 0$, we can conclude the contradiction $\frac{1}{\|A^{-1}\|} \leq \|A - B\|$ (to the assumption). $\square$

Problem 2.4. Show that a strictly row diagonally dominant matrix is invertible.

Solution. Let A be a strictly row diagonally dominant matrix. If $A = L + D + U$ is the decomposition of A , then the diagonal matrix D consists of non zero entries in the diagonal and hence invertible. Since $A = D \times D^{-1}A$, it is sufficient to prove that $D^{-1}A$ is invertible. Note that I is invertible matrix with $\|I^{-1}\|_\infty = 1$. Thus if we can show that $\|I - D^{-1}A\|_\infty < 1$, then by the application of the previous theorem it follows that $D^{-1}A$ is invertible. But it is easy to see that $\|I - D^{-1}A\|_\infty = \mu < 1$.

2.17. Ill-conditioned matrix

We first solve the following system of two linear equations in two unknowns.

$$\begin{aligned}x_1 + 3x_2 &= 19 \\ 2.5x_1 + 7.857x_2 &= 47.499\end{aligned}$$

The exact solution to this system is $x_1 = -3$ and $x_2 = 7$. But if we round off 47.499 by 47.500, then the solution changes drastically to $x_1 = 19$ and $x_2 = 0$. Or if we round off 7.857 by 7.86, the solution changes to $x_1 = 98.17$ and $x_2 = -26.39$.

We observe that a small change in the coefficient matrix A or the constant vector b leads to a large change in the solution vector. Such system is called ill-conditioned, otherwise the system is called well-conditioned.

2.18. Small change in b vector

We want to solve

$$Ax = b \tag{2.30}$$

Suppose the change δb in b leads to change δx in solution vector so that $A(x + \delta x) = Ax + A\delta x = b + \delta b$. This implies $A\delta x = \delta b$, or $\delta x = A^{-1}\delta b$. Thus

$$\|\delta x\| = \|A^{-1}\delta b\| \leq \|A^{-1}\| \|\delta b\| \tag{2.31}$$

Above equation shows that the δx is controlled if $\|A^{-1}\|$ is controlled. Further since

$$\|b\| = \|Ax\| \leq \|A\| \|x\|, \tag{2.32}$$

we can control the relative error in solution vector $\|\delta x\|/\|x\|$. Using (2.31),

$$\frac{\|\delta x\|}{\|x\|} \leq \|A^{-1}\| \times \|\delta b\| \frac{1}{\|x\|} \leq \|A^{-1}\| \times \|\delta b\| \frac{\|A\|}{\|b\|} = (\|A^{-1}\| \times \|A\|) \frac{\|\delta b\|}{\|b\|}. \tag{2.33}$$

Thus we see that the relative error in x is controlled by relative error in b if one has control over the quantity $(\|A^{-1}\|_{\infty} \|A\|_{\infty})$, which is known as **condition number** of matrix A .

Exercise 2.11. Show that for any invertible matrix A the condition number is always greater or equal to 1.

3 The Eigenvalue Problem

Let B be a square matrix. A number λ (real or complex) is said to be an eigenvalue of the matrix B if there exists a nonzero vector y such that

$$By = \lambda y.$$

This is equivalent to say that $(B - \lambda I)y = 0$ for some non-zero vector y ,

$\Leftrightarrow$ the null space of $(B - \lambda I)$ is not equal to $\{0\}$,

$\Leftrightarrow$ the dimension of the null space of $(B - \lambda I)$ is greater than or equal to 1,

$\Leftrightarrow (B - \lambda I)$ is singular,

$\Leftrightarrow$ determinant of $(B - \lambda I)$ is zero.

Clearly if y is an eigenvector for the matrix B corresponding to eigenvalue λ , then so is αy for any nonzero α

Note that any matrix B of order $m \times n$ represents a linear transformation from an n -dimensional vector space to an m -dimensional vector space with respect to some fixed ordered basis of these vector spaces. It can be proved that the linear map defined by the matrix B as discussed is a continuous map from n -dimensional vector space to an m -dimensional vector space.

Moreover, any square matrix of order $n \times n$ can be viewed as a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^n$ with respect to the standard basis.

The importance of eigenvector is that the image of an eigenvector of a square matrix is again a vector in the same direction but scaled by a factor of its eigenvalue.

Moreover, if there is a basis of eigenvectors say $\{v_1, v_2, \dots, v_n\}$, with corresponding eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$, then image of any vector is completely known. Suppose if $z \in \mathbb{R}^n$, then there are scalars $\alpha_1, \alpha_2, \dots, \alpha_n$ such that

$$z = \alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n, \quad (3.1)$$

so that

$$Bz = \alpha_1 Bv_1 + \alpha_2 Bv_2 + \dots + \alpha_n Bv_n = \alpha_1 \lambda_1 v_1 + \alpha_2 \lambda_2 v_2 + \dots + \alpha_n \lambda_n v_n.$$

In this case the matrix of linear transformation with respect to the basis $\{v_1, v_2, \dots, v_n\}$ turns out to be diagonal with diagonal entries as $\lambda_1, \lambda_2, \dots, \lambda_n$ and we say that the matrix B is diagonalizable.

Exercise 3.1. Show that if a matrix B is diagonalizable, with eigenvalues $|\lambda_1| \geq |\lambda_2| \geq \dots \geq |\lambda_n|$, then the image of the unit ball $\{x : \|x\| \leq 1\}$ is contained in a ball of radius $|\lambda_1|$ with center at origin.

This exercise shows the importance of eigenvalue of largest magnitude.

3.1. The Power Method

This method is useful to find the dominant eigenvalue among a collection of eigenvalues of a matrix and an eigenvector corresponding to the dominant eigenvalue. Let $\lambda_1, \lambda_2, \dots, \lambda_m$ be a set of eigenvalues of a square matrix of order $n \times n$, with corresponding eigenvectors $v_1, v_2, \dots, v_m$, $m \leq n$ such that $z = \alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_m v_m$, with $\alpha_1 \neq 0$ and $|\lambda_1| \geq |\lambda_2| \geq |\lambda_3| \geq \dots \geq |\lambda_m|$.

Thus,

$$B^k z = \alpha_1 \lambda_1^k v_1 + \alpha_2 \lambda_2^k v_2 + \dots + \alpha_m \lambda_m^k v_m. \quad (3.2)$$

Note that if u is a vector such that $\langle v_1, u \rangle \neq 0$, then $\langle z, u \rangle \neq 0$ and

$$\frac{\langle B^{k+1} z, u \rangle}{\langle B^k z, u \rangle} = \lambda_1 \frac{\alpha_1 \langle v_1, u \rangle + \alpha_2 (\lambda_2/\lambda_1)^{k+1} \langle v_2, u \rangle + \dots + \alpha_m (\lambda_m/\lambda_1)^{k+1} \langle v_m, u \rangle}{\alpha_1 \langle v_1, u \rangle + \alpha_2 (\lambda_2/\lambda_1)^k \langle v_2, u \rangle + \dots + \alpha_m (\lambda_m/\lambda_1)^k \langle v_m, u \rangle}. \quad (3.3)$$

So that in the limiting case

$$\lim_{k \rightarrow \infty} \frac{\langle B^{k+1} z, u \rangle}{\langle B^k z, u \rangle} = \lambda_1. \quad (3.4)$$

Moreover, from (3.2) $\lambda_1^{-k} B^k z = \alpha_1 v_1 + \alpha_2 (\lambda_2/\lambda_1)^k v_2 + \dots + \alpha_m (\lambda_m/\lambda_1)^k v_m$. Thus

$$\lim_{k \rightarrow \infty} \lambda_1^{-k} B^k z = \lim_{k \rightarrow \infty} (\lambda_1^{-1} B)^k z = \alpha_1 v_1. \quad (3.5)$$

Thus by (3.4), we can first find the largest eigenvalue and then by (3.2) the eigenvector, corresponding to the largest eigenvalue, involved in the representation of z . Note that the eigenvector $\lambda_1 v_1$ is not necessarily of unit length.

Problem 3.1. Use power method to find the largest eigenvalue and the corresponding eigenvector of the matrix $\begin{pmatrix} 3 & -1 \\ 2 & 0 \end{pmatrix}$ with the initial vector z as $\begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and $u = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

Solution. Clearly, since $z = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ is not the eigenvalue of the matrix $B = \begin{pmatrix} 3 & -1 \\ 2 & 0 \end{pmatrix}$. So we can infer that if there is a basis of eigenvectors, z is a linear combination of both of them. Further since $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is not an eigenvector, its inner product with both the eigenvectors has to be non-zero and hence we can use this vector as u vector. Now

$$\begin{aligned} Bz &= \begin{pmatrix} 3 & -1 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 8 \\ 6 \end{pmatrix} \Rightarrow \langle B^1 z, u \rangle = 8, \\ B^2 z &= BBz = \begin{pmatrix} 3 & -1 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} 8 \\ 6 \end{pmatrix} = \begin{pmatrix} 18 \\ 16 \end{pmatrix} \Rightarrow \langle B^2 z, u \rangle = 18, \\ B^3 z &= BB^2 z = \begin{pmatrix} 3 & -1 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} 18 \\ 16 \end{pmatrix} = \begin{pmatrix} 38 \\ 36 \end{pmatrix} \Rightarrow \langle B^3 z, u \rangle = 38, \\ B^4 z &= BB^3 z = \begin{pmatrix} 3 & -1 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} 38 \\ 36 \end{pmatrix} = \begin{pmatrix} 78 \\ 76 \end{pmatrix} \Rightarrow \langle B^4 z, u \rangle = 78, \\ B^5 z &= BB^4 z = \begin{pmatrix} 3 & -1 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} 78 \\ 76 \end{pmatrix} = \begin{pmatrix} 158 \\ 156 \end{pmatrix} \Rightarrow \langle B^5 z, u \rangle = 158, \\ B^6 z &= BB^5 z = \begin{pmatrix} 3 & -1 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} 158 \\ 156 \end{pmatrix} = \begin{pmatrix} 318 \\ 316 \end{pmatrix} \Rightarrow \langle B^6 z, u \rangle = 318, \\ B^7 z &= BB^6 z = \begin{pmatrix} 3 & -1 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} 318 \\ 316 \end{pmatrix} = \begin{pmatrix} 638 \\ 636 \end{pmatrix} \Rightarrow \langle B^7 z, u \rangle = 638, \\ B^8 z &= BB^7 z = \begin{pmatrix} 3 & -1 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} 638 \\ 636 \end{pmatrix} = \begin{pmatrix} 1278 \\ 1276 \end{pmatrix} \Rightarrow \langle B^8 z, u \rangle = 1278. \end{aligned}$$

Thus first initial terms of the sequence $\left\{ \frac{\langle B^{k+1} z, u \rangle}{\langle B^k z, u \rangle} \right\}$ are $\frac{18}{8} = 2.25$, $\frac{38}{18} = 2.1111$, $\frac{78}{38} = 2.0526$, $\frac{158}{78} = 2.0256$, $\frac{318}{158} = 2.0126$, $\frac{638}{318} = 2.0062$, $\frac{1278}{638} = 2.0031$. Thus up to two decimal places largest eigenvalue is 2. Moreover first few terms of the sequence $\{\lambda_1^{-k} B^k z\}$ are $[3, 2]^t$, $[4.5, 4]^t$, $[4.75, 4.5]^t$, $[4.875, 4.75]^t$, $[4.9375, 4.875]^t$, $[4.96875, 4.9375]^t$, $[4.984375, 4.96875]^t$, $[4.9921875, 4.984375]^t$. Thus the eigenvector correct up to one decimal place is $[5, 5]^t$. Note that $\begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \end{pmatrix} + \begin{pmatrix} -2 \\ -4 \end{pmatrix}$, where $\begin{pmatrix} -2 \\ -4 \end{pmatrix}$ should be the eigenvector corresponding to the other eigenvalue. And hence other eigenvalue is 1.

3.2. QR Decomposition

Let $A = [a_1, a_2, \dots, a_n]$ be a non-singular square matrix of order $n \times n$ such that $a_1, a_2, \dots, a_n$ are the column vectors of A . We apply Gram-Schmidt Process to find an orthonormal basis from the basis $\{a_1, a_2, \dots, a_n\}$. For this we first consider a unit vector in the direction of a_1

as $e_1 = \frac{a_1}{\|a_1\|}$. Next we search for a vector u_2 in the perpendicular direction of a_1 such that $\text{span}\{a_1, a_2\} = \text{span}\{e_1, u_2\}$, this can be obtained by subtracting from a_2 the projection of a_2 in the direction of a_1 , that is, $\frac{\langle a_2, a_1 \rangle}{\|a_1\|^2} a_1$. Thus $u_2 = a_2 - \langle a_2, e_1 \rangle e_1$. Consider unit vector in the direction of u_2 , that is, $e_2 = \frac{u_2}{\|u_2\|}$. Using $\langle a_2, e_2 \rangle = \langle u_2, e_2 \rangle$, we have $\langle a_2, e_2 \rangle e_2 = \langle u_2, e_2 \rangle e_2 = \|u_2\| e_2 = u_2$. Similarly we define $e_3, e_4, \dots, e_n$ and get

$$\begin{aligned} a_1 &= \langle a_1, e_1 \rangle e_1, \\ a_2 &= \langle a_2, e_1 \rangle e_1 + \langle a_2, e_2 \rangle e_2 \\ \text{In general, } e_k &= \sum_{j=1}^k \langle a_k, e_j \rangle e_j. \end{aligned}$$

Thus if we consider

$$Q = [e_1, e_2, \dots, e_n]^t, \quad \text{and} \quad R = \begin{bmatrix} \langle a_1, e_1 \rangle & \langle a_2, e_1 \rangle & \dots & \langle a_n, e_1 \rangle \\ 0 & \langle a_2, e_2 \rangle & \dots & \langle a_n, e_2 \rangle \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \langle a_n, e_n \rangle \end{bmatrix},$$

then $A = QR$.

Note that the norm used here is $\|\cdot\|_2$, which is compatible with inner product. Moreover, it can also be shown that QR decomposition of a non-singular square matrix is unique.

Problem 3.2. Find the QR decomposition of the matrix $\begin{bmatrix} -3 & -5 & -8 \\ 6 & 4 & 1 \\ -6 & 2 & 5 \end{bmatrix}$.

Solution. Note that $a_1 = [-3, 6, -6]^t$, $a_2 = [-5, 4, 2]^t$, and $a_3 = [-8, 1, 5]^t$. So that $\|a_1\| = \sqrt{9+36+36} = 9$ and $e_1 = [-1/3, 2/3, -2/3]^t$. Now $\langle a_1, e_1 \rangle = 9$, $\langle a_2, e_1 \rangle = 3$, $\langle a_3, e_1 \rangle = 0$ so that $u_2 = [-5, 4, 2]^t - 3[-1/3, 2/3, -2/3]^t = [-4, 2, 4]^t$. Thus $e_2 = [-2/3, 1/3, 2/3]^t$ and $\langle a_2, e_2 \rangle = 6$, $\langle a_3, e_2 \rangle = 9$. Now $u_3 = [-8, 1, 5]^t - 0e_1 - 9[-2/3, 1/3, 2/3]^t = [-2, -2, -1]^t$ so that $e_3 = [-2/3, -2/3, -1/3]^t$ and $\langle a_3, e_3 \rangle = 3$. Thus $Q = \frac{1}{3} \begin{bmatrix} -1 & -2 & -2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{bmatrix}$ and $R = \begin{bmatrix} 9 & 3 & 0 \\ 0 & 6 & 9 \\ 0 & 0 & 3 \end{bmatrix}$.

3.3. QR Algorithm

Let A_1 be a square matrix of order n with n distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ such that $|\lambda_1| > |\lambda_2| > \dots > |\lambda_n|$. Decompose A_1 as $A_1 = Q_1 R_1$, where R_1 is an upper triangular matrix and Q_1 is orthogonal matrix such that $Q_1^t = Q_1^{-1}$. Consider $A_2 = R_1 Q_1 = Q_1^{-1} Q_1 R_1 Q_1 = Q_1^{-1} A_1 Q_1$. Thus A_2 is similar to A_1 and hence the set of eigenvalues of A_2 is same as the set of eigenvalues of A_1 . Now if A_2 has the QR decomposition as $A_2 = Q_2 R_2$, we define $A_3 = R_2 Q_2$, which is again similar to A_2 and hence similar to A_1 . Thus we find a sequence of similar matrices $\{A_m\}$.

With certain conditions, the sequence of iterates $\{A_m\}$ converges to an upper triangular matrix.

Problem 3.3. Use QR algorithm to find the eigenvalues of the matrix $\begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}$.

Solution. Let $A_0 = \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}$. The QR factorisation of $A_0 = Q_1 R_1$ where $Q_1 = \frac{1}{\sqrt{13}} \begin{bmatrix} 3 & -2 \\ 2 & 3 \end{bmatrix}$ and $R_1 = \frac{1}{\sqrt{13}} \begin{bmatrix} 13 & 7 \\ 0 & 4 \end{bmatrix}$. Thus $A_2 = R_1 Q_1 = \frac{1}{13} \begin{bmatrix} 53 & -5 \\ 8 & 12 \end{bmatrix}$. This shows that the first approximation to the eigenvalues are $\frac{53}{13}, \frac{12}{13}$. To decompose A_2 as $Q_2 R_2$ we follow the same procedure to get $Q_2 = \frac{1}{13\sqrt{17}} \begin{bmatrix} 53 & -8 \\ 8 & 53 \end{bmatrix}$ and $R_2 = \frac{1}{\sqrt{17}} \begin{bmatrix} 17 & -1 \\ 0 & 4 \end{bmatrix}$. Now $A_3 = R_2 Q_2 = \frac{1}{221} \begin{bmatrix} 893 & -189 \\ 32 & 212 \end{bmatrix}$. Thus second approximation to the eigenvalues turns out to be $\frac{893}{221}$ and $\frac{212}{221}$. We can proceed further to find the next approximation.

3.4. Location of Eigenvalues

If λ is an eigenvalue of a square matrix B with eigenvector v , then $Bv = \lambda v$, or $\|Bv\| = |\lambda|\|v\|$ and hence

$$|\lambda| = \frac{\|Bv\|}{\|v\|} \leq \frac{\|B\| \times \|v\|}{\|v\|} = \|B\|.$$

Note that this is true for all possible matrix norms, that is, $\|\cdot\|_1$, $\|\cdot\|_2$ and $\|\cdot\|_\infty$. Thus

$$|\lambda| \leq \min\{\|B\|_1, \|B\|_2, \|B\|_\infty\}$$

Theorem 3.1. (Gershgorin's Theorem) *Let A be a square matrix B of order $n \times n$. Each eigenvalue λ of B satisfies*

$$|a_{ii} - \lambda| \leq \sum_{j=1, j \neq i}^n |a_{ij}|, \quad (3.6)$$

at least for some $1 \leq i \leq n$.

Proof. Let λ be the eigenvalue of the matrix B and v be an eigenvector corresponding to this eigenvalue so that $Av = \lambda v$. If $\|v\|_\infty = |v_k|$, then $|v_k| \geq |v_i|$ for all $1 \leq i \leq n$. Now since $\lambda v_k - a_{kk}v_k = a_{k1}v_1 + a_{k2}v_2 + \dots + a_{k,k-1}v_{k-1} + a_{k,k+1}v_{k+1} + \dots + a_{kn}v_n$, we have

$$\begin{aligned} |\lambda - a_{kk}| &= \left| a_{k1} \frac{v_1}{v_k} + a_{k2} \frac{v_2}{v_k} + \dots + a_{k,k-1} \frac{v_{k-1}}{v_k} + a_{k,k+1} \frac{v_{k+1}}{v_k} + \dots + a_{kn} \frac{v_n}{v_k} \right| \\ &\leq |a_{k1}| \frac{|v_1|}{|v_k|} + |a_{k2}| \frac{|v_2|}{|v_k|} + \dots + |a_{k,k-1}| \frac{|v_{k-1}|}{|v_k|} + |a_{k,k+1}| \frac{|v_{k+1}|}{|v_k|} + \dots + |a_{kn}| \frac{|v_n|}{|v_k|} \\ &\leq |a_{k1}| + |a_{k2}| + \dots + |a_{k,k-1}| + |a_{k,k+1}| + \dots + |a_{kn}|. \end{aligned}$$

Thus for $i = k$ the inequality (3.6) holds. $\square$

Remark 3.1. Since the set of eigenvalues of a square matrix A is as of its transpose A^t , one can apply the Gershgorin's theorem to A^t to conclude

$$|a_{ii} - \lambda| \leq \sum_{j=1, j \neq i}^n |a_{ji}|.$$

Problem 3.4. Use Gerschgorin's theorem to find the location of eigenvalues of the matrix

$$\begin{pmatrix} 1 & 0 & -1 \\ 1 & -2 & 1 \\ 2 & -1 & -1 \end{pmatrix}.$$

Solution. Let λ be eigenvalue of the given matrix. According to Gerschgorin's theorem the λ has to satisfy at least one of the following conditions. $|\lambda - 1| \leq 1$, $|\lambda + 2| \leq 1 + 1$ and $|\lambda + 1| \leq 2 + 1$. Thus all the eigenvalues of the matrix lie within the union of these three disks. Further if we apply Gershgorin's theorem to transpose of the given matrix, then λ should lie within the union of the disks $|\lambda - 1| \leq 1 + 2$, $|\lambda + 2| \leq 1$, and $|\lambda + 1| \leq 1 + 1$. Thus finally, we conclude that all the eigenvalues should lie within the intersection of these two unions.

4 Interpolation by polynomials

Suppose we want to know about the value of a function (or some of its derivative) at a particular point. But only information we know about the function is that we know its value (or the value of its derivatives) at certain other points. It might be some arbitrary continuous function and we want to approximate the function by some polynomial of least degree, which agrees with the known data about the function. The reason of approximating the given functions by a least degree polynomial is that we can compute the value of a polynomial at a point just by using basic operations (addition, subtraction and multiplication) of computer.

In this chapter our aim is to find an approximating polynomial to a function, whose value or the value of its certain derivatives is known at some points. These points are called nodes. To ease the computations we always try to find the minimal degree polynomial, which satisfies the given data. Such approximating polynomials are called interpolating polynomials.

Example 4.1. The polynomial $P(x) = x - f(x_0)$ is an interpolating polynomial for some function $f(x)$, because $P(x)$ agrees with $f(x)$ at the node $x = x_0$.

Example 4.2. If $f(x) = \cos x$, then the polynomial $P(x) = 1$ not only agrees with $f(x)$ at $x = 0$, but also the first derivative of $P(x)$ agrees with the first derivatives of $f(x)$ at $x = 0$. Thus $P(x) = 1$ is an interpolating polynomial for $\cos x$ with node $x = 0$.

Example 4.3. If $f(x) = \cos x$, then the polynomial $P(x) = 1 - \frac{x^2}{2}$ not only agrees with $f(x)$ at $x = 0$, but also the first and second derivatives of $P(x)$ agree with the first and second derivatives of $f(x)$ respectively at $x = 0$. Thus $1 - \frac{x^2}{2}$ is also an interpolating polynomial for $\cos x$.

Example 4.4. If $f(x) = \sin x$, then the polynomial $P(x) = x$ not only agrees with $f(x)$ at $x = 0$, but also the first and second derivative of $P(x)$ agrees with the respective derivatives of $f(x)$ at $x = 0$. Thus $P(x) = x$ is an interpolating polynomial for $\sin x$.

But the third order derivative of $P(x) = x$ does not coincide with the third order derivative of the function $\sin x$! If we consider $P(x) = x - \frac{x^3}{3!}$, then $P(x)$ agrees with $\sin x$ at $x = 0$ along with its first three derivatives.

Problem 4.1. Find a polynomial, which coincide with $|x|$ at the points $-1, 0$ and 1 .

It means we need to find a polynomial P , which should coincide with the function $|x|$ at the points $-1, 0$ and 1 . So we must have

$$P(-1) = |-1| = 1, P(0) = |0| = 0, P(1) = |1| = 1 \quad (4.1)$$

Now since there are three distinct points, we will try to find a two degree polynomial, because the degree of freedom of two degree polynomial is three. (The vector space of at most two degree polynomial is three dimensional and $\{1, x, x^2\}$ is a basis for this vector space.)

Let $P(x) = a + bx + cx^2$ be a polynomial, which passes through the points $(-1, 1), (0, 0), (1, 1)$. Then we must have

$$\begin{aligned} a + b(0) + c(0) &= 0 \\ a + b(-1) + c(-1)^2 &= 1 \\ a + b(1) + c(1)^2 &= 1 \end{aligned} \quad (4.2)$$

Remark 4.1. Here we need to solve a system of three linear equations in three unknowns. If we would have approximated it by a polynomial of degree less than two, then the system of three linear equations in two unknowns might not have any solution. And similarly if we would have approximated it by a four degree polynomial, then this system of linear equations might have infinite number of solutions. But because of increased degree of polynomial, computations will be difficult.

From (4.2), we must have

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & -1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \quad (4.3)$$

Since the $\det \left(\begin{bmatrix} 1 & 0 & 0 \\ 1 & -1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \right) = -2$, and the $\text{Adj} \begin{bmatrix} 1 & 0 & 0 \\ 1 & -1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 1 & -1 \\ 2 & -1 & -1 \end{bmatrix}$. The inverse of the matrix is given by

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & -1 & 1 \\ 1 & 1 & 1 \end{bmatrix}^{-1} = -\frac{1}{2} \begin{bmatrix} -2 & 0 & 0 \\ 0 & 1 & -1 \\ 2 & -1 & -1 \end{bmatrix}. \quad (4.4)$$

And hence

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = -\frac{1}{2} \begin{bmatrix} -2 & 0 & 0 \\ 0 & 1 & -1 \\ 2 & -1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}. \quad (4.5)$$

Thus the two degree polynomial which coincide with $|x|$ at $(-1, 0, 1)$ is given by x^2 .

Remark 4.2. In general there are large number of data points with numerically rich data.

- We need to compute the inverse of a large matrix.
- The solution of the system of linear equations may be far from that of exact solution due to round off errors.

Suppose we know the value of a continuous function at two points x_0, x_1 , that is, we know $f(x_0), f(x_1)$.

Question 4.1. Can we find a polynomial, which satisfies $P(x_0) = f(x_0), P(x_1) = f(x_1)$?

And the answer is very easy that we learned in our 10th standard. We find the equation of a line passing through $(x_0, f(x_0))$ and $(x_1, f(x_1))$, because the equation of line is a polynomial of degree one. I am sure that you know many versions of the formula of finding the equation of a line passing through two points. But we try to find this in some different way.

$$P(x) = ax + b; f(x_0) = ax_0 + b; f(x_1) = ax_1 + b;$$

or

$$P(x)(-1) + ax + (1)b = 0; f(x_0)(-1) + ax_0 + (1)b = 0; f(x_1)(-1) + ax_1 + (1)b = 0;$$

Now if you recall your matrix representations of system of linear equations from Linear Algebra course, we can write above system of equations as

$$\begin{bmatrix} P(x) & x & 1 \\ f(x_0) & x_0 & 1 \\ f(x_1) & x_1 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad (4.6)$$

Now you know that if for any particular value of x say x' the 3×3 matrix $\begin{bmatrix} P(x') & x' & 1 \\ f(x_0) & x_0 & 1 \\ f(x_1) & x_1 & 1 \end{bmatrix}$ is invertible, then by multiplying the inverse of this matrix on both sides of the equation (4.6), we get a contradiction that the matrix $\begin{bmatrix} -1 \\ a \\ b \end{bmatrix}$ is a null matrix. This implies that the matrix $\begin{bmatrix} P(x) & x & 1 \\ f(x_0) & x_0 & 1 \\ f(x_1) & x_1 & 1 \end{bmatrix}$

is a singular matrix for every value of x . And hence, the determinant of this matrix is zero for all values of x . Then by simplifying the determinant $\begin{vmatrix} P(x) & x & 1 \\ f(x_0) & x_0 & 1 \\ f(x_1) & x_1 & 1 \end{vmatrix}$ with respect to first column, we get

$$P(x)(x_0 - x_1) + f(x_0)(x_1 - x) + f(x_1)(x - x_0) = 0.$$

Rewriting it we get

$$P(x) = \frac{(x - x_1)}{(x_0 - x_1)} f(x_0) + \frac{(x - x_0)}{(x_1 - x_0)} f(x_1). \quad (4.7)$$

Observation 4.1. Now if we want to find a polynomial $P(x)$, which agrees with a function $f(x)$ at three distinct points x_0, x_1, x_2 , that is, $P(x)$ satisfies $P(x_0) = f(x_0)$, $P(x_1) = f(x_1)$ and $P(x_2) = f(x_2)$, and if we consider the two degree polynomial $P(x) = a + bx + cx^2 + dx^3$ and proceed similar to above case then we will come to a conclusion that the determinant of the matrix

$\begin{bmatrix} P(x) & x^2 & x & 1 \\ f(x_0) & x_0^2 & x_0 & 1 \\ f(x_1) & x_1^2 & x_1 & 1 \\ f(x_2) & x_2^2 & x_2 & 1 \end{bmatrix}$ is equal to zero for all values of x in the interval. By simplification with respect to first column we get.

$$P(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} f(x_0) + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} f(x_1) + \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} f(x_2). \quad (4.8)$$

Now if we write

$$\begin{aligned} L_0(x) &= \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)}, \\ L_1(x) &= \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)}, \\ L_2(x) &= \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)}, \end{aligned}$$

then we observe the following properties of these functions.

- $L_0 + L_1 + L_2 = 1$,
- $L_j(x_i) = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$,
- The degree of each L_j is two, and
- $P(x) = L_0(x)f(x_0) + L_1(x)f(x_1) + L_2(x)f(x_2)$.

The polynomials L_j 's are called Lagrange's fundamental polynomial. And the corresponding polynomial $P(x)$ is called Lagrange's polynomial.

Problem 4.2. Let $[a, b]$ is a given interval and f is some unknown continuous function defined on $[a, b]$, if we know the the value of the function at $n + 1$ distinct points $a = x_0 < x_1 < \dots < x_n = b$, that is we know $f(x_i)$ at $i = 0, \dots, n$. Can we find a polynomial

$$P(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n, \quad (4.9)$$

such that $P(x_i) = f(x_i)$ for $i = 0, \dots, n$?

4.1. Lagrange Polynomial

Using the similar procedure, as in the above two examples, J.L. Lagrange found a polynomial, usually denote it $L(x)$, as

$$L(x) = \sum_{j=0}^n \left(\prod_{i=0, i \neq j}^n \frac{(x - x_i)}{(x_j - x_i)} \right) f(x_j) = \sum_{j=0}^n L_j(x) f(x_j), \quad (4.10)$$

where $L_j(x)$ is j th fundamental polynomial of degree n and is given by

$$L_j(x) = \prod_{i=0, i \neq j}^n \frac{(x - x_i)}{(x_j - x_i)}. \quad (4.11)$$

These L_j 's satisfy the following properties.

- If we write $\omega(x) = \prod_{j=0}^n (x - x_j)$, then Lagrange's fundamental polynomial can also be expressed as

$$L_j(x) = \frac{\omega(x)}{(x - x_j)\omega'(x_j)}. \quad (4.12)$$

- $\sum_{j=0}^n L_j(x) = 1$,
- $L_j(x_k) = \delta_{jk} = \begin{cases} 1 & \text{if } j = k \\ 0 & \text{if } j \neq k \end{cases}$,
- the degree of each L_j is n , and
- $L(x) = \sum_{j=0}^n L_j(x) f(x_j)$. And
- the degree of $L(x)$ is at most n .

Remark 4.3. There are examples of data in which $L(x)$ is of degree strictly less than n .

Remark 4.4. We can use the the following steps to find Lagrange polynomial for the given data.

- First we compute Lagrange's fundamental polynomial L_j for each j .
- Multiply each $L_j(x)$ with the corresponding functional value $f(x_j)$.
- Then sum these multiplications obtained in step b) over all j 's.

Remark 4.5. If we draw the graphs of both $f(x)$ and $P(x)$, then it is clear from the conditions that these two graphs intersect each other at least at $n + 1$ distinct points namely $(x_0, f(x_0)), (x_1, f(x_1)), \dots, (x_n, f(x_n))$. In other words $P(x)$ interpolates $f(x)$ at $n + 1$ points.

Example 4.5. The Lagrange's polynomial is also an interpolating polynomial, because it matches with function at $n + 1$ distinct points in the interval. And can give some approximating value for the function at any other point in the interval with some error.

We can also think of finding a polynomial P such that certain derivatives of P coincide with the same order derivatives of the given function f at some points in the interval.

Example 4.6. If $f(x) = e^x$, then the polynomial

- $P_0(x) = 1$ is an interpolating polynomial for e^x , because $P_0(x) = 1$ agrees with the function at $x = 0$.
- $P_1(x) = 1 + x$ is an interpolating polynomial for e^x , because $P_1(x)$ and its first derivative agree with the function e^x at $x = 0$.

- $P_2(x) = 1 + x + \frac{x^2}{2}$ is an interpolating polynomial for e^x , because $P_2(x)$ and its first and second order derivatives agree with the function e^x at $x = 0$.
- $P_3(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!}$ is an interpolating polynomial for e^x , because $P_3(x)$ and its first three derivatives agree with the function e^x and derivatives respectively at $x = 0$.

Example 4.7. The Taylor's polynomial of degree n of the function $f(x)$ around the point x_0 is given by

$$P_n(x) = f(x_0) + (x - x_0)f'(x_0) + \frac{1}{2!}(x - x_0)^2 f''(x_0) + \dots + \frac{1}{n!}(x - x_0)^n f^n(x_0). \quad (4.13)$$

Then P_n agrees with $f(x)$ at $x = x_0$ along with its first n derivatives. Therefore, P_n is also an interpolating polynomial for the function $f(x)$.

Remark 4.6. In all above examples, to find the interpolating polynomial $P(x)$, we do not need the full information about the function. We only need the the specific data about the function. In Example (4.4), we can also pose the problem in a different way, without saying any thing about *sine* function as follows.

- Find a polynomial $P(x)$, which satisfy the following $P(0) = 0$, $P'(0) = 1$, $P''(0) = 0$, and $P'''(0) = -1$, or
- Find a polynomial $P(x)$, which agree with the function $f(x)$ along the data $f(0) = 0$, $f'(0) = 1$, $f''(0) = 0$, and $f'''(0) = -1$.

Note that this $f(x)$ need not be $\sin x$, for example it might be $x - \frac{x^3}{3!} + x^{10}$.

Example 4.8. The polynomial

$$P(x) = \frac{(x - x_1)}{(x_0 - x_1)}f(x_0) + \frac{(x - x_0)}{(x_1 - x_0)}f(x_1)$$

is an interpolating polynomial for $f(x)$, because $P(x)$ coincide with $f(x)$ at the points x_0, x_1 .

On the other hand if we want to find an interpolating polynomial for the data $f(x_0) = f(x_1) = c$, then above formula yields $P(x)$ to be a constant polynomial c . But the polynomial $P(x) = (x - x_1)(x - x_2) + c$ also satisfies the data. It means that the interpolating polynomial for a given data is not necessarily unique.

Question 4.2. Since the interpolating polynomial is not unique, the question arises that by which interpolating polynomial we should approximate the function.

In above example there are two interpolating polynomial for the given data, one is a constant polynomial and other is of degree two. Note that the calculations with the constant polynomial will be easier than that of two degree polynomial.

Remark 4.7. In general we need to deal with a large and numerically rich data, so to ease the computations it will be better to approximate by an interpolating polynomial of minimal degree.

Remark 4.8. In problem (4.1) we have interpolated a data with three distinct points by a two degree polynomial and by construction it is clear that this is the only polynomial of at most degree two, which can interpolate the given data. If we change the functional value as constant 1 at each of the above three points $(-1, 0, 1)$, and follow the same procedure as above, we get the interpolating polynomial as constant polynomial 1, which will be again unique by construction.

Question 4.3. If we are given the functional value at $(n + 1)$ distinct points, then by the answer of Problem (4.2) there exists a Lagrange's polynomial of at most degree n , which interpolates the given data. Now the question is that wether there are any other interpolating polynomials (for the same data) of degree at most n . Or in other words, wether the Lagrange's polynomial is unique interpolating polynomial of degree at most n for given data of $n + 1$ distinct points.

Answer to this is **Yes**.

Theorem 4.1. (Uniqueness of Interpolating Polynomial) *If we know the functional value of a real valued function f at $n + 1$ distinct points $x_0, x_1, \dots, x_n$, then there exists exactly one polynomial of degree at most n , which interpolates f at $x_0, x_1, \dots, x_n$.*

Clearly Lagrange's polynomial (4.10) is of degree at most n and interpolates the given data. Only thing we need to show is that it is unique interpolating polynomial of degree at most n . We will prove it in two ways.

Proof. First, without loss of generality we consider that $x_0 < x_1 < \dots < x_{n-1} < x_n$. Let $P(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$ be a polynomial, which satisfies the following data.

$$P(x_0) = f(x_0), P(x_1) = f(x_1), \dots, P(x_n) = f(x_n). \quad (4.14)$$

Then we must have

$$\begin{aligned} a_0 + a_1x_0 + a_2x_0^2 + \dots + a_nx_0^n &= f(x_0) \\ a_0 + a_1x_1 + a_2x_1^2 + \dots + a_nx_1^n &= f(x_1) \\ \dots & \\ a_0 + a_1x_n + a_2x_n^2 + \dots + a_nx_n^n &= f(x_n) \end{aligned} \quad (4.15)$$

Or in matrix form,

$$\begin{bmatrix} 1 & x_0 & x_0^2 & \dots & x_0^n \\ 1 & x_1 & x_1^2 & \dots & x_1^n \\ \vdots & \vdots & \vdots & \dots & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^n \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} f(x_0) \\ f(x_1) \\ \vdots \\ f(x_n) \end{bmatrix} \quad (4.16)$$

This a system of $(n + 1)$ linear equations in $(n + 1)$ unknowns $a_0, a_1, \dots, a_n$. This system will have a unique solution if determinant of following matrix (known as Vandermonde determinant) is non zero.

$$\Delta = \begin{vmatrix} 1 & x_0 & x_0^2 & \dots & x_0^n \\ 1 & x_1 & x_1^2 & \dots & x_1^n \\ \vdots & \vdots & \vdots & \dots & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^n \end{vmatrix} \neq 0 \quad (4.17)$$

But we know that $\Delta = \prod_{0 \leq i < j \leq n} (x_i - x_j) \neq 0$. So all the coefficients of $P(x)$ are uniquely determined and hence $P(x)$ itself. $\square$

For the second proof we need the following lemma.

Lemma 4.2. *Let f be a real valued function on real line. Further if f is n times differentiable function and has $(n + 1)$ zeros, then n_{th} derivative of the function f^n has at least one real zero.*

Above lemma can be proved by repeated application of Rolle's Theorem. We need the following corollary to this lemma for our second proof.

Corollary 4.3. *A non zero polynomial of degree n cannot have $n + 1$ distinct zeros.*

Now it is worth to note that even if we rearrange the order of data points, the interpolating polynomial will not change and hence a_n , the coefficient of x^n in interpolating polynomial, will remain the same. This fact is also justified by the expression on R.H.S. of (4.21). Thus we can say that the n_{th} divided difference $f[x_0, x_1, x_2, \dots, x_n]$ is dependent on the set of node points $\{x_0, x_1, x_2, \dots, x_n\}$ but independent of the order of node points.

In the following we aim to justify the name divided difference. It is easy to see from (4.20) that

$$a_1 = \frac{f(x_1) - f(x_0)}{(x_1 - x_0)}.$$

Here, a_1 is obtained by dividing the difference of functional values by the difference of points and hence it is known as Newton's first divided difference of $f(x)$ relative to x_0, x_1 and denoted by $f[x_0, x_1]$. In general Newton's first divided difference of $f(x)$ relative to x_j, x_k , ($j \neq k$) is defined as

$$f[x_j, x_k] = \frac{f(x_j) - f(x_k)}{(x_j - x_k)}. \quad (4.22)$$

It is clear from the definition that $f[x_j, x_k]$ is independent of the order of j and k and depends only on the set $\{j, k\}$. Using these expressions for a_0 and a_1 , we can compute a_2 from first three equations of (4.20).

$$a_2 = f[x_0, x_1, x_2] = \frac{f(x_0)}{(x_0 - x_1)(x_0 - x_2)} + \frac{f(x_1)}{(x_1 - x_0)(x_1 - x_2)} + \frac{f(x_2)}{(x_2 - x_0)(x_2 - x_1)}. \quad (4.23)$$

Clearly, $f[x_0, x_1, x_2]$ is independent of order of x_0, x_1, x_2 . In fact one can write $f[x_0, x_1, x_2]$ as the fraction of the difference of certain first divided differences by difference of certain points as following.

Exercise 4.1. Show that

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{(x_2 - x_0)} = \frac{f[x_1, x_2] - f[x_2, x_0]}{(x_1 - x_0)} = \frac{f[x_0, x_2] - f[x_0, x_1]}{(x_2 - x_1)}. \quad (4.24)$$

Hence $f[x_0, x_1, x_2]$ is known as Newton's second divided difference of $f(x)$ relative to $\{x_0, x_1, x_2\}$. In general we define Newton's second divided difference of $f(x)$ relative to $\{x_i, x_j, x_k\}$ as

$$f[x_i, x_j, x_k] = \frac{f[x_j, x_k] - f[x_i, x_j]}{(x_k - x_i)}. \quad (4.25)$$

Moreover we can proceed in the similar way and can see that $f[x_0, x_1, x_2, \dots, x_{k-1}, x_k]$, the Newton's k_{th} divided difference of $f(x)$ relative to $\{x_0, x_1, \dots, x_{k-1}, x_k\}$, can be written as the fraction of difference of certain $(k-1)_{th}$ divided difference by the difference of certain point values as follows.

$$f[x_0, x_1, x_2, \dots, x_{k-1}, x_k] = \frac{f[x_1, x_2, \dots, x_{k-1}, x_k] - f[x_0, x_1, x_2, \dots, x_{k-1}]}{(x_k - x_0)}. \quad (4.26)$$

It can be checked that

$$f[x_0, x_1, x_2, \dots, x_{k-1}, x_k] = \sum_{i=0}^k \frac{f(x_i)}{\prod_{j \neq i} (x_i - x_j)}, \quad k = 3, \dots, n. \quad (4.27)$$

In general we define Newton's k_{th} divided difference of $f(x)$ relative to $\{x_{i_0}, x_{i_1}, \dots, x_{i_{k-1}}, x_{i_k}\}$ recursively as follows

$$f[x_{i_0}, x_{i_1}, x_{i_2}, \dots, x_{i_{k-1}}, x_{i_k}] = \frac{f[x_{i_1}, x_{i_2}, \dots, x_{i_{k-1}}, x_{i_k}] - f[x_{i_0}, x_{i_1}, x_{i_2}, \dots, x_{i_{k-1}}]}{(x_{i_k} - x_{i_0})}. \quad (4.28)$$

Now we can rewrite the Newton's interpolating polynomial by plugging the the expressions for a_k 's as Newton's divided difference in (4.19).

$$\begin{aligned} P(x) &= f(x_0) + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) + \dots \\ &\quad \dots + f[x_0, x_1, \dots, x_n](x - x_0)(x - x_1) \dots (x - x_{n-1}), \\ &= f(x_0) + \sum_{i=1}^n (x - x_0)(x - x_1) \dots (x - x_{i-1}) f[x_0, x_1, \dots, x_i] \end{aligned} \quad (4.29)$$

Thus to find the polynomial $P(x)$ completely, we need to find $f[x_0, x_1]$, $f[x_0, x_1, x_2]$, $\dots$, $f[x_0, x_1, \dots, x_n]$, that is, the divided difference of f of all order from 1 to n . One can directly calculate $f[x_0, x_1]$, but to find $f[x_0, x_1, x_2]$ we also need to calculate first difference of f relative to points x_1, x_2 , that is, $f[x_1, x_2]$. And further to find $f[x_0, x_1, x_2, x_3]$, we need to find $f[x_1, x_2, x_3]$, that is, we need an extra computation of $f[x_2, x_3]$. If we proceed in this way, we need to compute the following table

$$\left[\begin{array}{cccccc} x & f & 1_{st} f[x_i, x_{i+1}] & 2_{nd} f[x_i, x_{i+1}, x_{i+2}] & \dots & n_{th} f[x_0, \dots, x_n] \\ x_0 & f(x_0) & f[x_0, x_1] = \frac{f(x_1) - f(x_0)}{(x_1 - x_0)} & f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{(x_2 - x_0)} & \dots & f[x_0, \dots, x_n] \\ x_1 & f(x_1) & f[x_1, x_2] = \frac{f(x_2) - f(x_1)}{(x_2 - x_1)} & f[x_1, x_2, x_3] = \frac{f[x_2, x_3] - f[x_1, x_2]}{(x_3 - x_1)} & \dots & \dots \\ x_2 & f(x_2) & f[x_2, x_3] = \frac{f(x_3) - f(x_2)}{(x_3 - x_2)} & f[x_2, x_3, x_4] = \frac{f[x_3, x_4] - f[x_2, x_3]}{(x_4 - x_2)} & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots \\ x_{n-1} & f(x_{n-1}) & f[x_{n-1}, x_n] & \dots & \dots & \dots \\ x_n & f(x_n) & \dots & \dots & \dots & \dots \end{array} \right].$$

This table will be of upper triangular form and the coefficients $a_0 = f(x_0)$, $a_1 = f[x_0, x_1]$, $a_2 = f[x_0, x_1, x_2]$, $\dots$, $a_n = f[x_0, x_1, \dots, x_n]$ are the entries of the first row. So by directly substituting the values of the coefficients in (4.19), we can obtain Newton's divided difference interpolating polynomial.

Problem 4.3. Find Newton's interpolating polynomial for the following data.

$$f(0) = 3, f(1) = 5, f(3) = 33, f(4) = 67, f(5) = 133.$$

First we form the divided difference table for the above data.

$$\begin{array}{cccccc} x & f & 1_{st} & 2_{nd} & 3_{rd} & 4_{th} \\ 0 & 3 & \frac{3-3}{(1-0)} = 0 & \frac{12-0}{(3-0)} = 4 & \frac{8-4}{(4-0)} = 1 & \frac{1-1}{(5-0)} = 0 \\ 1 & 3 & \frac{27-3}{(3-1)} = 12 & \frac{36-12}{(4-1)} = 8 & \frac{12-8}{(5-1)} = 1 & \dots \\ 3 & 27 & \frac{63-27}{(4-3)} = 36 & \frac{60-36}{(5-3)} = 12 & \dots & \dots \\ 4 & 63 & \frac{123-63}{(5-4)} = 60 & \dots & \dots & \dots \\ 5 & 123 & \dots & \dots & \dots & \dots \end{array} \quad (4.30)$$

Now we substitute the corresponding values divided differences in formula for Newton's interpolating polynomial of degree four.

$$\begin{aligned} P(x) &= f(0) + f[0, 1](x - 0) + f[0, 1, 3](x - 0)(x - 1) + \\ &\quad f[0, 1, 3, 4](x - 0)(x - 1)(x - 3) + f[0, 1, 3, 4, 5](x - 0)(x - 1)(x - 3)(x - 4) \\ P(x) &= 3 + 0(x - 0) + 4(x - 0)(x - 1) + 1(x - 0)(x - 1)(x - 3) + 0(x - 0)(x - 1)(x - 3)(x - 4) \\ P(x) &= 3 + 4x(x - 1) + x(x - 1)(x - 3) \end{aligned}$$

4.3. Divided difference at variable point (Functional value in terms of divided difference.)

In the following, we will try to find the divided difference of the function at some arbitrary point of the interval. And we will see that the functional value at some given point can be expressed in terms of divided difference at that point.

$$\begin{aligned} f[x, x_0] &= \frac{f(x_0) - f(x)}{(x_0 - x)}. \\ \implies f(x) &= f(x_0) + (x - x_0)f[x, x_0]. \end{aligned} \quad (4.31)$$

Further,

$$\begin{aligned} f[x, x_0, x_1] &= \frac{f[x_0, x_1] - f[x, x_0]}{(x_1 - x)}. \\ \implies f[x, x_0] &= f[x_0, x_1] + (x - x_1)f[x, x_0, x_1]. \end{aligned} \quad (4.32)$$

Substituting (4.32) in (4.31), we get

$$f(x) = f(x_0) + (x - x_0)f[x_0, x_1] + (x - x_0)(x - x_1)f[x, x_0, x_1]. \quad (4.33)$$

Continuing this procedure up to n^{th} divided difference, we get inductively that

$$\begin{aligned} f(x) &= f(x_0) + (x - x_0)f[x_0, x_1] + (x - x_0)(x - x_1)f[x_0, x_1, x_2] + \dots \\ &\quad + (x - x_0)(x - x_1) \dots (x - x_{n-1})f[x, x_0, x_1, \dots, x_{n-1}]. \end{aligned} \quad (4.34)$$

But

$$\begin{aligned} f[x, x_0, x_1, \dots, x_n] &= \frac{f[x_0, x_1, x_2, \dots, x_n] - f[x, x_0, x_1, \dots, x_{n-1}]}{(x_n - x)} \\ \implies f[x, x_0, x_1, \dots, x_{n-1}] &= f[x_0, x_1, x_2, \dots, x_n] + (x - x_n)f[x, x_0, x_1, \dots, x_n]. \end{aligned}$$

Thus from (4.34), we get

$$\begin{aligned} f(x) &= f(x_0) + (x - x_0)f[x_0, x_1] + (x - x_0)(x - x_1)f[x_0, x_1, x_2] \\ &\quad + \dots + (x - x_0)(x - x_1) \dots (x - x_{n-1})f[x_0, x_1, x_2, \dots, x_n] \\ &\quad + (x - x_0)(x - x_1) \dots (x - x_n)f[x, x_0, x_1, \dots, x_n]. \end{aligned} \quad (4.35)$$

Using the expression for Newton's polynomial form (4.29), we get from (4.35)

$$f(x) = P(x) + (x - x_0)(x - x_1) \dots (x - x_n)f[x, x_0, x_1, \dots, x_n]. \quad (4.36)$$

4.4. Properties of divided difference

As we saw that first divided difference relative to points x_0, x_1 is defined as $f[x_0, x_1] = \frac{f(x_1) - f(x_0)}{(x_1 - x_0)}$. But this expression reminds us (**mean value theorem**) that if we also assume the differentiability of the function, our first divided difference $f[x_0, x_1]$ is nothing but the derivative of the function f at some point in the smallest open interval containing both x_0, x_1 .

Question 4.4. Can we say that second divided difference is related to second derivative of the function?

If yes, then how?

If we assume that the function is two times differentiable and $x_0 < x_1 < x_2$, then we write

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{(x_2 - x_0)} = \frac{f'(\xi_1) - f'(\xi_2)}{(x_2 - x_0)} = \frac{(\xi_1 - \xi_0)}{(x_2 - x_0)} \frac{f'(\xi_1) - f'(\xi_2)}{(\xi_1 - \xi_0)} = \frac{(\xi_1 - \xi_0)}{(x_2 - x_0)} f''(\zeta), \quad (4.37)$$

where $\xi_0 \in (x_0, x_1), \xi_1 \in (x_1, x_2)$ and $\zeta \in (\xi_0, \xi_1)$.

Observation 4.2. Here we note that if ξ_0, ξ_1 are the middle points of the intervals (x_0, x_1) and (x_1, x_2) respectively, then $f[x_0, x_1, x_2] = \frac{1}{2}f''(\zeta)$.

We will return to this and prove a relation between divided difference and derivative of the function in the Remark 4.12 of Theorem 4.4. Now we see some other properties of divided differences.

Using Exercise 4.1, one can see that Newton's second divided difference is independent of the order of node points and depends only on the set of all node points. If we consider the function $\omega(x) = \prod_{j=0}^n (x - x_j)$, then we can see that

$$f[x_0, x_1, x_2, \dots, x_n] = \sum_{i=0}^n \frac{f(x_i)}{\omega'(x_i)}. \quad (4.38)$$

We know that $f[x, x_0] = \frac{f(x_0) - f(x)}{(x_0 - x)}$. This shows that $f[x, x_0]$ is a continuous function of x for all $(x_0 \neq)x \in [a, b]$, possibly undefined at x_0 . But if we consider f to be differentiable function at x_0 , then we can define

$$f[x_0, x_0] = \lim_{x \rightarrow x_0} f[x, x_0] = \lim_{x \rightarrow x_0} \frac{f(x_0) - f(x)}{(x_0 - x)} = \frac{d}{dx} f|_{x=x_0} = f'(x_0). \quad (4.39)$$

Further, if we assume that f is once differentiable in whole interval (a, b) , then it is clear by mean value theorem that $f[x_0, x] = \frac{f(x) - f(x_0)}{(x - x_0)} = f'(\tilde{x})$ for some $\tilde{x} \in (\min\{x_0, x\}, \max\{x_0, x\})$. Moreover,

$$\begin{aligned} \frac{d}{dx} f[x_0, x_1, \dots, x_n, x] &= \lim_{h \rightarrow 0} \frac{f[x_0, x_1, \dots, x_n, x+h] - f[x_0, x_1, \dots, x_n, x]}{h}, \\ &= \lim_{h \rightarrow 0} \frac{f[x_0, x_1, \dots, x_n, x+h] - f[x, x_0, x_1, \dots, x_n]}{(x+h) - x}, \\ &= \lim_{h \rightarrow 0} f[x, x_0, x_1, \dots, x_n, x+h], \\ &= f[x, x_0, x_1, \dots, x_n, x], \\ &= f[x_0, x_1, \dots, x_n, x, x]. \end{aligned} \quad (4.40)$$

Problem 4.4. Use induction to prove that

$$\frac{d^k}{dx^k} f[x_0, x_1, \dots, x_n, x] = k! f[x_0, x_1, \dots, x_n, \overbrace{x, x, \dots, x}^{(k+1) \text{ times}}]. \quad (4.41)$$

Now we further assume f to be second differentiable function and define

$$\begin{aligned} f[x_0, x_0, x_0] &= \lim_{x_1 \rightarrow x_0, x_2 \rightarrow x_0} f[x_0, x_1, x_2], \\ &= \lim_{x_1 - x_0 \rightarrow 0, x_2 - x_0 \rightarrow 0} f[x_0, x_1, x_2], \\ &= \lim_{h \rightarrow 0, k \rightarrow 0} f[x_0, x_0 + h, x_0 + k], \\ &= \lim_{h \rightarrow 0, k \rightarrow 0} \frac{f[x_0 + h, x_0 + k] - f[x_0, x_0 + h]}{(x_0 + k) - x_0}, \\ &= \lim_{k \rightarrow 0} \frac{f[x_0, x_0 + k] - f[x_0, x_0]}{k}, \\ &= \lim_{k \rightarrow 0} \frac{\frac{f(x_0 + k) - f(x_0)}{k} - f'(x_0)}{k}, \\ &= \lim_{k \rightarrow 0} \frac{f(x_0 + k) - f(x_0) - kf'(x_0)}{k^2}. \end{aligned}$$

Using L'Hospital's Rule, we have now

$$\begin{aligned}
f[x_0, x_0, x_0] &= \lim_{k \rightarrow 0} \frac{f(x_0 + k) - f(x_0) - kf'(x_0)}{k^2} \\
&= \lim_{k \rightarrow 0} \frac{f'(x_0 + k) - f'(x_0)}{2k} \\
&= \frac{1}{2} f''(x_0).
\end{aligned} \tag{4.42}$$

Remark 4.10. Let f be n times differentiable, then in the similar manner one can define $f[x_0, x_0, \dots, x_0] = \lim_{h_i \rightarrow 0} f[x_0, x_0 + h_1, \dots, x_0 + h_n]$ and using induction one can obtain that

$$f[\overbrace{x_0, x_0, \dots, x_0}^{(n+1) \text{ times}}] = \frac{1}{n!} f^n(x_0). \tag{4.43}$$

4.5. Error in Interpolation

Till now we approximated a continuous real valued function f defined on interval $[a, b]$ by the interpolating polynomials P_n at node points $x_0 = a, x_1, \dots, x_n = b$. But interpolating polynomials P_n does not necessarily match with f at some arbitrary point of the interval other than node points. And we expect some difference between $P_n(\tilde{x})$ and $f(\tilde{x})$ at some point $\tilde{x} \in [a, b], \tilde{x} \neq x_i$. This difference is known as error in interpolation. Since by Theorem 4.1 there is unique interpolation polynomial P_n of degree at most n , we can define error function as

$$E_n f(x) = f(x) - P_n(x). \tag{4.44}$$

Remark 4.11. We note that Newton's divided difference interpolating polynomial $P_n(x)$ given by (4.29) interpolates the function $f(x)$ at $n + 1$ distinct points and the degree of $P_n(x)$ is at most n , but Lagrange's polynomial $L(x)$ given by (4.11) also interpolates the same data so by Theorem 4.1 both $L(x)$ and $P_n(x)$ are the same polynomial written in two different ways. Therefore, for a given continuous function f the error function defined by (4.44) depends only on the set of node points. From expression (4.36) it is clear that by interpolating the function $f(x)$ by a Newton's divided difference interpolating polynomial $P_n(x)$, the error $E_n f(x)$ can be expressed a

$$E_n f(x) = f(x) - P_n(x) = (x - x_0)(x - x_1) \dots (x - x_n) f[x, x_0, x_1, \dots, x_n]. \tag{4.45}$$

Question 4.5. Can we estimate this error?

Theorem 4.4. Let f be $n + 1$ times differentiable function defined on interval $[a, b]$. Let P_n be the unique interpolating polynomial of degree at most n , interpolating f at $n + 1$ distinct points $x_0, x_1, x_2, \dots, x_n$. Then the error in interpolation at some point $x_i \neq x \in [a, b]$ is given by

$$E_n f(x) = (x - x_0)(x - x_1) \dots (x - x_n) \frac{f^{n+1}(\xi)}{(n+1)!}. \tag{4.46}$$

Where ξ is some point in the interval $(\min\{x, x_0, x_1, \dots, x_n\}, \max\{x, x_0, x_1, \dots, x_n\})$ and depends upon x .

Proof. Let x be some point in the interval other than node points. Let $\phi(t)$ be a function defined on interval $[a, b]$ by

$$\phi(t) = f(t) - P_n(t) - (t - x_0)(t - x_1) \dots (t - x_n)K, \tag{4.47}$$

where K is a constant determined by the equation $\phi(x) = 0$, that is,

$$f(x) - P_n(x) + (x - x_0)(x - x_1) \dots (x - x_n)K = 0. \tag{4.48}$$

It is clear that $\phi(x_i) = 0$ for all $i = 0, 1, 2, \dots, n$. Thus function ϕ has $n + 2$ zeros say $\{x, x_0, x_1, \dots, x_n\}$. Hence its $(n + 1)_{th}$ derivative has at least one zero at some point ξ in the interval $(\min\{x, x_0, x_1, \dots, x_n\}, \max\{x, x_0, x_1, \dots, x_n\})$, that is, $\phi^{n+1}(\xi) = 0$. Therefore,

$$0 = \phi^{n+1}(\xi) = f^{n+1}(\xi) - 0 - (n + 1)!K. \quad (4.49)$$

This implies that $K = \frac{1}{(n+1)!}f^{n+1}(\xi)$. And hence from (4.47) and (4.48),

$$E_n f(x) = f(x) - P_n(x) = (x - x_0)(x - x_1) \dots (x - x_n) \frac{1}{(n + 1)!} f^{n+1}(\xi). \quad (4.50)$$

□

Remark 4.12. Equating (4.45) and (4.46), we get

$$f[x, x_0, x_1, \dots, x_n] = \frac{1}{(n + 1)!} f^{n+1}(\xi). \quad (4.51)$$

Where $\xi \in (\min\{x, x_0, x_1, \dots, x_n\}, \max\{x, x_0, x_1, \dots, x_n\})$.

Corollary 4.5. If $P_m(x) = a_0 + a_1x + a_2x^2 + \dots + a_mx^m$ is of degree m , then m_{th} divided difference of P_m is a_m and $(m + 1)_{th}$ divided difference is zero.

Proof. For $f(x) = P_m(x)$, we have from (4.51) that $P_m[x, x_0, x_1, \dots, x_n] = \frac{1}{(n+1)!}P_m^{n+1}(\xi)$. Thus $P_m[x, x_0, x_1, \dots, x_m] = \frac{1}{m+1!}P_m^{m+1}(\xi) = 0$, or $P_m[x_0, x_1, \dots, x_m, x_{m+1}] = 0$. Similarly $P_m[x, x_0, x_1, \dots, x_{m-1}] = \frac{1}{m!}P_m^m(\xi) = a_m$, or $P_m[x_0, x_1, \dots, x_{m-1}, x_m] = a_m$. □

4.6. Bound on the Error in Interpolation

We note from (4.46) that the error in interpolation by n degree polynomial is given by

$$E_n f(x) = (x - x_0)(x - x_1) \dots (x - x_n) \frac{f^{n+1}(\xi)}{(n + 1)!} = \frac{f^{n+1}(\xi)}{(n + 1)!} \omega(x). \quad (4.52)$$

If we denote by

$$M_{n+1} = \max_{x \in [a, b]} |f^{n+1}(x)|, \quad (4.53)$$

then the error is bounded by

$$|E_n f(x)| \leq \frac{M_{n+1}}{(n + 1)!} |(x - x_0)(x - x_1) \dots (x - x_n)|. \quad (4.54)$$

If we some how know the bound for $|(x - x_0)(x - x_1) \dots (x - x_n)|$ on $[a, b]$ or the $\max_{x \in [a, b]} |(x - x_0)(x - x_1) \dots (x - x_n)|$, we can find the uniform bound for the error function. Certainly, the maxima and minima is attained by the continuous function $\omega(x) = (x - x_0)(x - x_1) \dots (x - x_n)$ on some points of the closed interval $[a, b]$. And we can bound $E_n f(x)$ by $\frac{M_{n+1}}{(n+1)!} \max\{|minimal|, |maximal|\}$. Since points of maxima or minima are also the points of local maxima and minima, so if we know all the points of local minima and maxima, that is, the critical points, we only need to compute the modulus of functional values at these critical points and find the maximum of those values. But we know that the critical points for some differentiable function are given by the roots of the derivative of the function. In our case critical points are given by the roots of the derivative of the polynomial $\omega(x)$, that is, by the solution of the equation $\omega'(x) = 0$. But to find these roots might not be an easy task if there is a large number of node points.

Problem 4.5. Find the bound on the error by interpolating a polynomial of five degree $x^5 - x^3 + 1$ by Lagrange's interpolating polynomial of degree two at the points 0, 2, 3.

Solution. We will find the solution in following steps.

- Since the number of node points are two, we need to interpolate by a polynomial of degree three.
- Since we are looking to interpolate the function with the given node points, so the interval of our concern is the smallest interval containing all the nodes. In our case this is $[0, 3]$.
- Since the error function in our case is given by $E(x) = (x-0)(x-2)(x-3)\frac{1}{(2+1)!}f^{2+1}(\xi)$, to find the bound on this it is enough to find the bound on $(x-0)(x-2)(x-3)$ and $f'''(x)$ on the interval $[0, 3]$.
- We have $f'''(x) = 60x^2 - 6$. Since the derivative of $60x^2 - 6$ is non-negative on the interval $[0, 3]$, f''' is non increasing function. Thus the maximum of $|f'''(x)|$ on the interval is possible at the end points of interval. So clearly this maximum is 534.
- Further we also need to compute the maximum of $|x^3 - 5x^2 + 6x|$ on $[0, 3]$. This can be attained either at the end point of the interval or at some local maxima or minima. But at end points, $\omega(x) = x^3 - 5x^2 + 6x$ is zero, so we only need to consider the critical points of $\omega(x)$ in the interval $[0, 3]$. These critical points can be obtained by solving $3x^2 - 10x + 6 = 0$. So points of extremum are $x_e = (5 \pm \sqrt{7})/3 = 2.54, .78$. And the local extremum values will be given by $\omega(x_e) = -.63, .21$. thus the maximum value of $|x^3 - 5x^2 + 6x|$ on $[0, 3]$ is .63.
- Thus,

$$\begin{aligned} |E(x)| &= |(x-0)(x-2)(x-3)\frac{1}{(2+1)!}f^{2+1}(\xi)| \\ &\leq |(x-0)(x-2)(x-3)| \frac{1}{(3)!} |f^3(\xi)| \\ &\leq .63 \times .16 \times 534 = 53.83 \end{aligned}$$

Remark 4.13. Thus we see in above problem that error in interpolation might be significantly large. But suppose if we have freedom to choose the position of node points, (not the number of nodes), wether we can control the error.

Question 4.6. More precisely, suppose we are given a smooth function f . And we aim to interpolate f by $n + 1$ node points. Further suppose that the choice of node points is in our hand. Can we choose these $n + 1$ nodes in such a way that the maximum error on the smallest interval containing all the nodes is controlled?

Suppose we assume that the node points are equally spaced, that is, $a = x_0, < x_1 = x_0 + h, < x_2 = x_0 + 2h, \dots < x_n = x_0 + nh = b$. And we know some how that $|f^{n+1}(x)| \leq M_{n+1}$. Then it is clear from from (4.54) that error function is bounded as follows.

$$|E_n f(x)| \leq \frac{M_{n+1}}{(n+1)!} |(x-x_0)(x-x_0-h)\dots(x-x_0-nh)|. \quad (4.55)$$

Clearly $\omega(x) = (x-x_0)(x-x_0-h)\dots(x-x_0-nh)$ is independent of the function. Here we will try to find a bound for $\omega(x) = (x-x_0)(x-x_0-h)\dots(x-x_0-nh)$ on the smallest interval containing all nodes, that is, $[x_0, x_0 + nh]$.

We will find the bound for $\omega(x)$ in three different cases of linear ($n = 1$), quadratic ($n = 2$), and cubic ($n = 3$) interpolation.

In case of linear interpolation it is easy to observe that the maximum value of $|(x-x_0)(x-x_1)|$ is obtained at $x = (x_0 + x_1)/2$ and is given by $(x_1 - x_0)^2/4$. And thus the bound for error in interpolation by a linear polynomial (line) is given by $M_2(x_1 - x_0)^2/8$, that is,

$$|E_1 f(x)| \leq M_2(x_1 - x_0)^2/8. \quad (4.56)$$

In case of quadratic interpolation we need to find the maximum value of $|\omega(x)| = |(x - x_0)(x - x_1)(x - x_2)|$. This maximum value can be obtained at one of the roots of quadratic polynomial $|\omega'(x)|$, which certainly has two real roots. But in case of equidistant points we need to find the maximum value of $|\omega(x)| = |(x - x_0)(x - x_0 - h)(x - x_0 - 2h)|$ in the interval $[x_0, x_0 + 2h]$. This extremum will be attained at some critical point given by $\omega'(x) = 0$ or $(x - x_0 - h)(x - x_0 - 2h) + (x - x_0)(x - x_0 - 2h) + (x - x_0)(x - x_0 - h) = 0$. For simplification we reparameterize the curve $\omega(x)$ with origin as the middle point of the interval $[x_0, x_0 + 2h]$ by assuming $x - x_0 = (t + 1)h$, then $x \in [x_0, x_0 + 2h] \Leftrightarrow t \in [-1, 1]$ and we need to solve $t(t - 1) + (t + 1)(t - 1) + (t + 1)t = 0$, that is, $3t^2 - 1 = 0$. Thus $t = \pm 1/\sqrt{3}$ or $x = x_0 + h \pm h/\sqrt{3}$. But at both of these values of x , $|\omega(x)| = 2h^3/\sqrt{27}$. Thus from (4.55), It is clear that

$$|E_2 f(x)| \leq \frac{M_3}{3!} 2h^3/\sqrt{27}. \quad (4.57)$$

Now we can bound $E_2 f$ by any given positive small number by choosing step size h accordingly small.

Similarly in cubic case if we know M_4 , then to bound $E_3 f$, we need to control $|\omega(x)| = |(x - x_0)(x - x_0 - h)(x - x_0 - 2h)(x - x_0 - 3h)|$. Thus we need to find the solution for $\omega'(x) = 0$. For simplification we reparameterize the curve $\omega(x)$ with origin as the center of the interval $[x_0, x_0 + 3h]$. Let $x - x_0 = (t + 3/2)h$, then $x \in [x_0, x_0 + 3h] \Leftrightarrow t \in [-3/2, 3/2]$ and we need to solve $d/dt[(t^2 - 9/4)(t^2 - 1/4)] = 0$, which gives $t = 0, \pm\sqrt{5}/2$. And the maximum value of $\omega(x)$ is 1. And hence

$$|E_3 f(x)| \leq \frac{M_4}{4!} h^4. \quad (4.58)$$

Problem 4.6. Determine the maximum step size that can be used in the tabulation of $f(x) = e^x$ in $[0, 1]$, so that the error for cubic interpolation will be less than 5×10^{-4} .

Solution. Since we want to approximate f by a three degree polynomial, we need to bound the error function $E_3 f$ using (4.58). For this we need to find M_4 , that is, maximum of the fourth derivative of $f(x) = e^x$ on the interval $[0, 1]$. But $f^{4th}(x) = e^x$, so $M_4 = e$. Since we want or error to be bounded by 5×10^{-4} , we need the step size h such that

$$|E_3 f(x)| \leq \frac{M_4}{4!} h^4 = (e/24)h^4 \leq 5 \times 10^{-4}.$$

Or we want

$$h^4 \leq (5 \times 24 \times 10^{-4})/e = .004414553294,$$

or,

$$h \leq (.004414553294)^{.25} = .25776366.$$

Observation 4.3. Suppose if we approximate the function $f(x) = x^3 - x^2 + 2$ at the nodes $-1, 0, 1$, the two degree polynomial, which satisfies $f(-1) = 2, f(0) = 2, f(1) = 2$, is given by $P(x) = -x^2 + x + 2$. If we draw the graphs of these functions, we observe that the graphs intersect each other at $x = 1$, but goes away rapidly as soon as they leave $x = 1$. And this is because they intersect perpendicularly at $x = 1 (P'(1) = -1, f'(1) = 1)$. So it is desirable to approximate the function with a polynomial, which not only agrees with the function at nodes but derivative of the polynomial also agrees with the derivative of the function at node points.

4.7. Hermite Interpolation

Problem 4.7. Find a polynomial $P(x)$ of minimal degree which agrees with the function f along the data

$$P(x_0) = f(x_0), P'(x_0) = f'(x_0), P(x_1) = f(x_1), P'(x_1) = f'(x_1). \quad (4.59)$$

Solution. Since there are four conditions, it is reasonable to consider the three degree polynomial $P(x) = a_0 + a_1x + a_2x^2 + a_3x^3$, in four unknown coefficients, (a_0, a_1, a_2, a_3) , which are to be determined by the given data (4.59). Since $P(x)$ satisfies (4.59), we have

$$\begin{aligned} a_0(1) + a_1x_0 + a_2x_0^2 + a_3x_0^3 &= f(x_0) \\ a_0(0) + a_1(1) + a_22x_0 + a_33x_0^2 &= f'(x_0) \\ a_0(1) + a_1x_1 + a_2x_1^2 + a_3x_1^3 &= f(x_1) \\ a_0(0) + a_1(1) + a_22x_1 + a_33x_1^2 &= f'(x_1) \end{aligned}$$

Or in matrix form,

$$\begin{bmatrix} 1 & x_0 & x_0^2 & x_0^3 \\ 0 & 1 & 2x_0 & 3x_0^2 \\ 1 & x_1 & x_1^2 & x_1^3 \\ 0 & 1 & 2x_1 & 3x_1^2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} f(x_0) \\ f'(x_0) \\ f(x_1) \\ f'(x_1) \end{bmatrix} \quad (4.60)$$

One can check that the $\det \begin{bmatrix} 1 & x_0 & x_0^2 & x_0^3 \\ 0 & 1 & 2x_0 & 3x_0^2 \\ 1 & x_1 & x_1^2 & x_1^3 \\ 0 & 1 & 2x_1 & 3x_1^2 \end{bmatrix} = (x_1 - x_0)^4 \neq 0$, so we will have a unique solution

to the above system of four equations. But to find solution is a tedious task. (So we leave it to very enthusiastic reader.)

Question 4.7. Can we find above polynomial in terms of Lagrange's fundamental polynomials? Can we find the values of $a_i, b_i, c_i, d_i, i = 0, 1, 2, \dots, n$, so that the polynomial

$$P(x) = \sum_{i=0}^n [(a_i x + b_i) L_i^2(x) f(x_i) + (c_i x + d_i) L_i^2(x) f'(x_i)], \quad (4.61)$$

satisfies $P(x_i) = f(x_i)$ and $P'(x_i) = f'(x_i)$ for all $i = 0, 1, 2, \dots, n$.

Problem 4.8. Suppose we denote

$$H_i(x) = (a_i x + b_i) L_i^2(x), \quad (4.62)$$

and

$$K_i(x) = (c_i x + d_i) L_i^2(x), \quad (4.63)$$

then find these a_i, b_i, c_i, d_i such that

$$H_i(x_j) = \delta_{ij}, K_i(x_j) = 0, H'_i(x_j) = 0, K'_i(x_j) = \delta_{ij}, \quad (4.64)$$

so that the polynomial

$$P(x) = \sum_{i=0}^n [H_i(x) f(x_i) + K_i(x) f'(x_i)], \quad (4.65)$$

automatically satisfies the conditions satisfies $P(x_i) = f(x_i)$ and $P'(x_i) = f'(x_i)$ for all $i = 0, 1, 2, \dots, n$.

Solution. We use the definition of H_i, K_i in (4.64) to obtain

$$(a_i x_i + b_i) L_i^2(x_i) = 1 \quad (4.66)$$

$$(a_i x_j + b_i) L_i^2(x_j) = 0 \quad (4.67)$$

$$a_i L_i^2(x_i) + (a_i x_i + b_i) 2L_i(x_i) L'_i(x_i) = 0 \quad (4.68)$$

$$a_i L_i^2(x_j) + (a_i x_j + b_i) 2L_i(x_j) L'_i(x_j) = 0 \quad (4.69)$$

$$(c_i x_i + d_i) L_i^2(x_i) = 0 \quad (4.70)$$

$$(c_i x_j + d_i) L_i^2(x_j) = 0 \quad (4.71)$$

$$c_i L_i^2(x_i) + (c_i x_i + d_i) 2L_i(x_i) L'_i(x_i) = 1 \quad (4.72)$$

$$c_i L_i^2(x_j) + (c_i x_j + d_i) 2L_i(x_j) L'_i(x_j) = 0 \quad (4.73)$$

Clearly (4.67),(4.69),(4.71),(4.73) are obviously satisfied because $L_i(x_j) = 0$. And to satisfy other conditions, we need

$$(a_i x_i + b_i) = 1 \quad (4.74)$$

$$a_i + (a_i x_i + b_i) 2L'_i(x_i) = 0 \quad (4.75)$$

$$(c_i x_i + d_i) = 0 \quad (4.76)$$

$$c_i + (c_i x_i + d_i) 2L'_i(x_i) = 1 \quad (4.77)$$

After solving them we get,

$$a_i = -2L'_i(x_i) \quad (4.78)$$

$$b_i = 1 + 2x_i L'_i(x_i) \quad (4.79)$$

$$c_i = 1 \quad (4.80)$$

$$d_i = -x_i \quad (4.81)$$

Thus $P(x)$ is given by

$$P(x) = \sum_{i=0}^n [1 - 2(x - x_i)L'_i(x_i)] L_i^2(x) f(x_i) + \sum_{i=0}^n (x - x_i) L_i^2(x) f'(x_i). \quad (4.82)$$

This polynomial is called as Hermite polynomial. Here we can write $L'_i(x_i)$ in terms of $\omega(x)$ as follows

$$L'_i(x_i) = \frac{\omega''(x_i)}{2\omega'(x_i)}. \quad (4.83)$$

Remark 4.14. In Hermite interpolation there are $n + 1$ nodes, and interpolating polynomial should satisfy $2(n + 1)$ equations. This suggests that our $P(x)$ should be of degree at most $2n + 1$. And clearly from (4.82), $P(x)$ satisfies this minimal degree criteria.

Theorem 4.6. (*Uniqueness of Hermite Interpolating Polynomial*) If we know the value of a real valued function f and its derivative f' at $n + 1$ distinct points $x_0 < x_1 < \dots < x_n$, then there exists exactly one polynomial of degree at most $2n + 1$, which satisfies the data $P(x_i) = f(x_i)$ and $P'(x_i) = f'(x_i)$ for all $i = 0, 1, 2, \dots, n$.

Proof. Clearly, the existence of such a polynomial is given by (4.82). We only need to prove the uniqueness.

Suppose if there is some polynomial $Q(x) \neq P(x)$ of degree at most $2n + 1$ such that

$$P(x_i) = f(x_i) = Q(x_i), P'(x_i) = f'(x_i) = Q'(x_i).$$

Then $\phi(x) = P(x) - Q(x) \neq 0$ is a non zero polynomial of degree at most $2n + 1$, such that $\phi(x_i) = 0$ and $\phi'(x_i) = 0$. But $\phi(x_i) = 0$ for all $i = 0, 1, 2, \dots, n$ implies that $\phi'(\xi_i) = 0$, for some $\xi_i \in (x_{i-1}, x_i)$ for all $i = 1, 2, \dots, n$. Thus we found n distinct zeros of ϕ' other than x'_i s. This shows that ϕ' has $2n + 1$ distinct zeros, which is a contradiction to the fact that $\phi'(x)$ is nonzero polynomial of degree at most $2n$ (see Corollary 4.3). We note that $\phi'(x)$ can not be a zero polynomial because $\phi(x)$ can not be a non zero constant polynomial as $\phi(x_i) = 0$. $\square$

We can also compute the error in Hermite interpolation.

Theorem 4.7. Let f be $2n + 2$ times differentiable real valued function defined on interval $[a, b]$. Let P_{2n+1} be the unique Hermite interpolating polynomial of degree at most $2n + 1$, satisfying

$$P_{2n+1}(x_i) = f(x_i), P'_{2n+1}(x_i) = f'(x_i), \quad i = 0, 1, 2, \dots, n. \quad (4.84)$$

Then the error in interpolation at some point $x_i \neq x \in [a, b]$ is given by

$$E_{2n+1}f(x) = (x - x_0)^2(x - x_1)^2 \dots (x - x_n)^2 \frac{1}{(2n+2)!} f^{2n+2}(\xi) = \frac{\omega^2(x)}{(2n+2)!} f^{2n+2}(\xi). \quad (4.85)$$

Where ξ is some point in the smallest interval I containing all the node points and depends upon x . And error bound on I is given by

$$|E_{2n+1}f(x)| \leq \max_{x \in I} |\omega^2(x)| \frac{M_{2n+2}}{(2n+2)!}. \quad (4.86)$$

Proof. Let x be some point in the interval other than node points. Let $\phi(t)$ be a function defined on interval $[a, b]$ by

$$\phi(t) = f(t) - P_{2n+1}(t) - (t - x_0)^2(t - x_1)^2 \dots (t - x_n)^2 K, \quad (4.87)$$

where K is a constant determined by the equation $\phi(x) = 0$, that is,

$$f(x) - P_{2n+1}(x) - (x - x_0)^2(x - x_1)^2 \dots (x - x_n)^2 K = 0. \quad (4.88)$$

It is clear that $\phi(x_i) = 0$ for all $i = 0, 1, 2, \dots, n$. Thus function ϕ has $n + 2$ zeros say $\{x, x_0, x_1, \dots, x_n\}$. Hence ϕ' must have at least $n + 1$ zeros at some points other than $\{x, x_0, x_1, \dots, x_n\}$. Moreover $\phi'(x_i) = 0$ for all $i = 0, 1, 2, \dots, n$. Thus ϕ' has at least $2n + 2$ distinct zeros. Hence $(2n + 1)_{th}$ derivative of ϕ' , that is, $\phi^{(2n+2)}$ must have at least one zero ξ in the interval I . Therefore,

$$0 = \phi^{2n+2}(\xi) = f^{2n+2}(\xi) - 0 - (2n + 2)!K. \quad (4.89)$$

This implies that $K = \frac{1}{2n+2!} f^{2n+2}(\xi)$. And hence from (4.87) and (4.88),

$$E_{2n+1}f(x) = f(x) - P_{2n+1}(x) = (x - x_0)^2(x - x_1)^2 \dots (x - x_n)^2 \frac{1}{(2n+2)!} f^{2n+2}(\xi). \quad (4.90)$$

Clearly (4.86) follows if we consider $M_{2n+2} = \max_{x \in I} |f^{2n+2}(x)|$. □

Problem 4.9. Find Lagrange's fundamental polynomial to interpolate the following data.

$$f(0) = 2, f'(0) = 0, f(1) = 2, f'(1) = 1.$$

Solution. Let $x_0 = 0, x_1 = 1$. And $L_0(x) = \frac{(x-x_1)}{(x_0-x_1)} = \frac{(x-1)}{(0-1)}$. Similarly $L_1(x) = \frac{(x-x_0)}{(x_1-x_0)} = \frac{(x-0)}{(1-0)}$. Thus, $L'_0(x_0) = L'_0(0) = -1$ and $L'_1(x_1) = L'_1(1) = 1$. Using,

$$P(x) = \sum_{i=0}^n [1 - 2(x - x_i)L'_i(x_i)] L_i^2(x) f(x_i) + \sum_{i=0}^n (x - x_i) L_i^2(x) f'(x_i).$$

$$\begin{aligned} P_3(x) &= \sum_{i=0}^1 [1 - 2(x - x_i)L'_i(x_i)] L_i^2(x) f(x_i) + \sum_{i=0}^1 (x - x_i) L_i^2(x) f'(x_i), \\ &= [1 - 2(x - 0)L'_0(0)] L_0^2(x) f(0) + [1 - 2(x - 1)L'_1(1)] L_1^2(x) f(1) \\ &\quad + (x - 0) L_0^2(x) f'(0) + (x - 1) L_1^2(x) f'(1) \\ &= [1 - 2x(-1)](1 - x)^2 2 + [1 - 2(x - 1)1]x^2 2 \\ &\quad + x(1 - x)^2(0) + (x - 1)x^2(1) \\ &= 2(1 + 2x)(x - 1)^2 + 2(3 - 2x)x^2 + (x - 1)x^2 \\ &= x^3 - x^2 + 2 \end{aligned}$$

4.8. Newton's method to find Hermite Interpolating Polynomial

We further consider Problem (4.59). And assume that the polynomial of degree three in the following form.

$$P(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + a_3(x - x_0)^2(x - x_1). \quad (4.91)$$

If this polynomial satisfy the data $P(x_0) = f(x_0), P'(x_0) = f'(x_0), P(x_1) = f(x_1), P'(x_1) = f'(x_1)$, then we must have

$$\begin{aligned} a_0 &= f(x_0) \\ a_1 &= f'(x_0) \\ a_0 + a_1(x_1 - x_0) + a_2(x_1 - x_0)^2 &= f(x_1) \\ a_1 + 2a_2(x_1 - x_0) + a_3(x_1 - x_0)^2 &= f'(x_1) \end{aligned}$$

If we assume that the function is smooth enough (as many times differentiable as we want), then $f'(x_0)$ can be written as $f[x_0, x_0]$, and $f'(x_1) = f[x_1, x_1]$. Hence, we have $f(x_0) + f[x_0, x_0](x_1 - x_0) + a_2(x_1 - x_0)^2 = f(x_1)$ or $f[x_0, x_0] + a_2(x_1 - x_0) = f[x_0, x_1]$, or $a_2 = f[x_0, x_0, x_1]$. Further from the last equation we have $a_1 + a_2(x_1 - x_0) + a_2(x_1 - x_0) + a_3(x_1 - x_0)^2 = f[x_1, x_1]$, or $f[x_0, x_1] + a_2(x_1 - x_0) + a_3(x_1 - x_0)^2 = f[x_1, x_1]$, or $f[x_0, x_0, x_1](x_1 - x_0) + a_3(x_1 - x_0)^2 = f[x_1, x_1] - f[x_0, x_1]$, or $f[x_0, x_0, x_1] + a_3(x_1 - x_0) = f[x_0, x_1, x_1]$, or $a_3 = f[x_0, x_0, x_1, x_1]$. Thus

$$P(x) = f(x_0) + f[x_0, x_0](x - x_0) + f[x_0, x_0, x_1](x - x_0)^2 + f[x_0, x_0, x_1, x_1](x - x_0)^2(x - x_1). \quad (4.92)$$

Remark 4.15. In general if there are $n + 1$ nodes one can compute **Newton's divided difference table with each node considered twice**, that is, total $2n + 2$ total number of nodes. And to compute the the first divided difference relative to repeated points x_i, x_i we directly use the data given to us because $f[x_i, x_i] = f'(x_i)$.

Example 4.9. We will use Newton's extended divided difference table to find a polynomial, which satisfies the following data in which at each node we need to match derivatives up to certain order.

$$P(0) = 1, P'(0) = 2, P''(0) = -2, P(1) = 3, P'(1) = 5, P''(1) = 18, P'''(1) = 60.$$

x	$f(x)$	$f[x, x]$	$f[x, x, x]$	$f[x, x, x, x]$
0	1	$f[0, 0] = f'(0) = 2$	$f[0, 0, 0] = f''(0)/2 = -1$	$f[0, 0, 0, 1] = 0 - (-1) = 1$
0	1	$f[0, 0] = f'(0) = 2$	$f[0, 0, 1] = 0$	$f[0, 0, 1, 1] = 3 - 0 = 3$
0	1	$f[0, 1] = (3 - 1)/(1 - 0) = 2$	$f[0, 1, 1] = 3$	$f[0, 1, 1, 1] = 9 - 3 = 6$
1	3	$f[1, 1] = f'(1) = 5$	$f[1, 1, 1] = f''(1)/2 = 9$	$f[1, 1, 1, 1] = f'''(1)/6 = 10$
1	3	$f[1, 1] = f'(1) = 5$	$f[1, 1, 1] = f''(1)/2 = 9$	...
1	3	$f[1, 1] = f'(1) = 5$	...	...
1	3	...	...	...

x	$f[x, x, x, x, x]$	$f[x, x, x, x, x, x]$	$f[x, x, x, x, x, x, x]$
0	$f[0, 0, 0, 1, 1] = 3 - 1 = 2$	$f[0, 0, 0, 1, 1, 1] = 1$	$f[0, 0, 0, 1, 1, 1, 1] = 0$
0	$f[0, 0, 1, 1, 1] = 6 - 3 = 3$	$f[0, 0, 1, 1, 1, 1] = 1$	...
0	$f[0, 1, 1, 1, 1] = 10 - 6 = 4$	...	...
1	...	...	...
1	...	...	...
1	...	...	...
1	...	...	...

$$P(x) = f(0) + f[0, 0]x + f[0, 0, 0]x^2 + f[0, 0, 0, 1]x^3 + f[0, 0, 0, 1, 1]x^3(x - 1) + f[0, 0, 0, 1, 1, 1]x^3(x - 1)^2 + f[0, 0, 0, 1, 1, 1, 1]x^3(x - 1)^3 = x^5 - x^2 + 2x + 1.$$

Remark 4.16. One can not use Newton's extended divided difference table if the value of certain order derivative is known at some node, but the value of lower order derivative is not known. We can not find a polynomial of degree three such that $P(0) = 1, P''(0) = 2, P''(1) = 0, P'''(1) = 1$.

5 Nonlinear Equation

In last section we learned to find the solution of a system of linear equations. But in practical problems it is also very important to find the value (or values) of an unknown satisfying certain nonlinear equation.

Question 5.1. Let f be a **nonlinear continuous** function defined on real line. Can we determine the values of x satisfying

$$f(x) = 0, \quad x \in \mathbb{R} \quad (5.1)$$

Remark 5.1. It might be easy to find the solution for $f(x) = 0$, if we know some how that this solution will lie in some particular interval. By using Intermediate value theorem one can find such an interval $[a, b]$ by determining two points $a, b \in \mathbb{R}$ such that $f(a)$ and $f(b)$ have different signs.

Now we will discuss some iterative methods to find such a solution.

5.1. Bisection Method

This method is based on the fact that if a continuous function f is changing the sign at two points, then f must have at least one root in the smallest interval containing these two points. And further subdivide this interval in two subintervals of equal length and search for the subinterval in which the root lies. This method can be described in following steps.

- First determine the interval $[a, b]$ in which the root lies. Which can be determined by finding two points a, b such that $f(a)f(b) < 0$. We consider middle point x_1 of $[a, b]$ as the first approximation to the root. If x_1 is the exact root then done or otherwise proceed to next step.
- Next choose the interval $[a_2, b_2]$ with one endpoint as x_1 and other endpoint as one of a or b according to $f(x_1)f(a) < 0$ or $f(x_1)f(b) < 0$ respectively. Now find second approximation x_2 as the middle point of $[a_2, b_2]$ such that $|x_2 - r| < \frac{|b_2 - a_2|}{2} = \frac{|b - a|}{4}$. If x_2 is the exact root then done or otherwise proceed to next step.
- In general find the k_{th} approximation x_k to root as the middle point of $[a_k, b_k]$. If x_k is the root, stop the process. And if not, find next interval $[a_{k+1}, b_{k+1}]$ such that one endpoint of this interval is x_k and other endpoint is one of a_k or b_k according to $f(x_k)f(a_k) < 0$ or $f(x_k)f(b_k) < 0$ respectively. Note that

$$|x_k - r| \leq \frac{|b_k - a_k|}{2} = \frac{|b - a|}{2^k}. \quad (5.2)$$

5.2. Algorithm

Bisect (f, a, b , root, ϵ, n)

1. $n = 1$
2. Define $c := \frac{a+b}{2}$;
3. If $b - c \leq \epsilon$, then accept root $:= c$, and exit;
4. If $\text{sgn}(f(b)) \cdot \text{sgn}(f(c)) \leq 0$, then $a := c$; otherwise, $b := c$
5. $n := n + 1$
6. Return to step 2.

5.3. Error analysis in bisection method

In bisection method we search for the location of the a root r of (5.1) lying in each interval $[a, b]$. The equation (5.2) shows that

$$|e_k| \leq \frac{(b-a)}{2^k}.$$

Using this inequality we can approach to the root as closer as we want by increasing the number of iterates.

5.4. The secant method

This method is based upon the linear approximation of the function. First we search two points near the root of the function, name these points as x_0, x_1 such that $f(x_1) \neq f(x_0)$. These points x_0, x_1 are first two approximations to the root. The line passing through two points $(x_0, f(x_0)), (x_1, f(x_1))$ on the graph of the function is called secant. We consider this secant as the approximating function to find the next iterate x_2 to root as the intersection point of this secant with the real line. One can draw a figure and check that the slope of this secant can be written in two different ways using similar triangles rule (assuming without loss of generality that $x_1 < x_0$ and $0 < f(x_1) < f(x_0)$).

$$\frac{f(x_0)}{x_0 - x_2} = \frac{f(x_0) - f(x_1)}{x_0 - x_1}. \quad (5.3)$$

After solving this we get

$$x_2 = x_1 - f(x_1) \frac{x_0 - x_1}{f(x_0) - f(x_1)}. \quad (5.4)$$

Now we will use x_1, x_2 to determine the secant as the line passing through $(x_1, f(x_1)), (x_2, f(x_2))$ on the graph of the original function and consider x_3 as the intersection of this secant with the real line. In general, we consider the secant passing through the graphical points $(x_{k-1}, f(x_{k-1})), (x_k, f(x_k))$ and the next approximation x_{k+1} as the intersection point of the secant with the real line. We have

$$x_{k+1} = x_k - f(x_k) \frac{x_{k-1} - x_k}{f(x_{k-1}) - f(x_k)}. \quad (5.5)$$

Remark 5.2. The sequence of iterates does not need to converge to root of the function. In fact it might also diverse to infinity.

Question 5.2. Then, why does one use secant method instead of bisection method, which gives the security of convergence.

5.5. Order of convergence and asymptotic error

Suppose the sequence of iterates $\{x_0, x_1, \dots\}$ converges to root r and for the sequence of errors $\{e_k = r - x_k\}$, there exists a number p and a constant $C \neq 0$ such that

$$\lim_{k \rightarrow \infty} \frac{|e_{k+1}|}{|e_k|^p} = C, \quad (5.6)$$

then p is called the order of convergence and C is called the asymptotic error.

5.6. Error analysis of the secant method

Since we do not know exact positioning of the root, our treatment to error will be different from bisection method, where we knew for sure the interval in which the root lies. Let us go back to the theme of secant method: linear approximation to the function at each new iterate. In fact for linear approximation with nodes as x_{k-1}, x_k , we can use (4.35) to write the expression for the function as

$$f(x) = f(x_k) + f[x_k, x_{k-1}](x - x_k) + f[x_k, x_{k-1}, x](x - x_k)(x - x_{k-1}). \quad (5.7)$$

If r is the root for the function, that $f(r) = 0$, we have from (5.7)

$$0 = f(x_k) + f[x_k, x_{k-1}](r - x_k) + f[x_k, x_{k-1}, r](r - x_k)(r - x_{k-1}).$$

This implies (assuming $f[x_k, x_{k-1}] \neq 0$)

$$r = x_k - \frac{f(x_k)}{f[x_k, x_{k-1}]} - \frac{f[x_k, x_{k-1}, r]}{f[x_k, x_{k-1}]}(r - x_k)(r - x_{k-1}).$$

Now using (5.5) we get

$$r = x_{k+1} - \frac{f[x_k, x_{k-1}, r]}{f[x_k, x_{k-1}]}(r - x_k)(r - x_{k-1}). \quad (5.8)$$

Further if $r - x_k = e_k$ denotes the error at k_{th} iterate, then we get from (5.8) that

$$e_{k+1} = -\frac{f[x_k, x_{k-1}, r]}{f[x_k, x_{k-1}]}e_k e_{k-1}. \quad (5.9)$$

To determine the order of convergence of the secant method we note that from (5.9)

$$|e_{k+1}| = c_k |e_k e_{k-1}|, \quad (5.10)$$

where

$$c_k = \left| \frac{f[x_k, x_{k-1}, r]}{f[x_k, x_{k-1}]} \right|. \quad (5.11)$$

Now if we assume that the sequence of iterates converges to the root r and the function is twice continuously differentiable, we have

$$\lim_{k \rightarrow \infty} c_k = |f''(r)/2f'(r)| := c. \quad (5.12)$$

Now using (5.10), we can write

$$\frac{|e_{k+1}|}{|e_k|^p} = c_k |e_k|^{1-p} |e_{k-1}| = c_k \left(\frac{|e_k|}{|e_{k-1}|^p} \right)^\alpha, \quad (5.13)$$

provided $\alpha = 1 - p$ and $\alpha p = -1$, that is, $p^2 - p - 1 = 0$. If we consider $y_k = |e_k|/|e_{k-1}|^p$, we have from (5.13)

$$y_{k+1} = c_k y_k^{-1/p} \quad (5.14)$$

Now if $y_k \rightarrow y$, then $y = cy^{-1/p}$ or $y = c^{1/p}$.

Note that the negative value of p is not considered because it leads to $\lim_{k \rightarrow \infty} |e_{k+1}|/|e_k|^p = c^{1/p}$, where $|e_k| \rightarrow 0$ implies $|e_{k+1}| \rightarrow \infty$ a contradiction to convergence, provided $f'(r) \neq 0$.

Thus the asymptotic error is $|f''(r)/2f'(r)|^{1/p}$ and order of convergence is $p = (1 + \sqrt{5})/2$.

5.7. Newton's Method

Newton's method of finding the root of the equation $f(x) = 0$ is based upon the approximation of the function with the tangent line near the root. Thus our function is assumed to be differentiable. Here we only need one point x_0 near the root as the initial approximation to begin with. To find next approximation we draw the tangent line at $(x_0, f(x_0))$ and consider the intersection point of this tangent line with the real line and name it as x_1 . Thus

$$f'(x_0) = \frac{f(x_0)}{x_0 - x_1}.$$

This gives

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}. \quad (5.15)$$

In general to find $(k+1)_{th}$ iterate, we use tangent line at $(x_k, f(x_k))$ as the approximation function and consider its root, that is, the intersection point with the real line. And we have

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}. \quad (5.16)$$

Remark 5.3. Newton's method also need not to converge.

5.8. Error analysis of Newton's method

One can write the function in the Newtonian form as

$$f(x) = f(x_k) + f[x_k, x_k](x - x_k) + f[x_k, x_k, x](x - x_k)^2. \quad (5.17)$$

If r is the root of the function, we have $0 = f(x_k) + f[x_k, x_k](r - x_k) + f[x_k, x_k, r](r - x_k)^2$. Or,

$$r = x_k - \frac{f(x_k)}{f[x_k, x_k]} - (r - x_k)^2 \frac{f[x_k, x_k, r]}{f[x_k, x_k]} = x_{k+1} - (r - x_k)^2 \frac{f[x_k, x_k, r]}{f[x_k, x_k]}. \quad (5.18)$$

Further if $e_k = r - x_k$ is the error at k_{th} stage, $e_{k+1} = -\frac{f[x_k, x_k, r]}{f[x_k, x_k]} e_k^2$, if f is twice differentiable at $x = r$, we have $\lim_{k \rightarrow \infty} \frac{f[x_k, x_k, r]}{f[x_k, x_k]} = \frac{f''(r)}{2f'(r)}$, hence

$$\lim_{k \rightarrow \infty} \frac{|e_{k+1}|}{|e_k|^2} = \frac{|f''(r)|}{2|f'(r)|}$$

Above equation shows that if the Newton's method converges, then order of convergence is 2 with asymptotic error as $|f''(r)/2f'(r)|$.

5.9. Fixed point iteration method

If $f(x) = 0 \Leftrightarrow x = g(x)$, then instead of finding roots of $f(x)$ we search for the fixed points of the function $g(x)$ (both are same). Consider some initial approximation to fixed point say x_0 to determine first approximation by the equation $x_1 = g(x_0)$ and in general

$$x_{n+1} = g(x_n) \quad \text{for all } n \in \mathbb{N}. \quad (5.19)$$

Theorem 5.1. If $g : [a, b] \rightarrow [a, b]$ is a continuous function, then there is at least one fixed point for g in $[a, b]$. If g is differentiable in (a, b) and

$$|g'(x)| \leq \alpha < 1 \text{ for all } x \in (a, b).$$

then there is exactly one fixed point of g in (a, b) . Further if $x_1 \in (a, b)$, then the sequence, defined by

$$x_n = g(x_{n-1}),$$

converges to the fixed point.

Proof. Since the function $g(x) - x$ changes sign in $[a, b]$, by Intermediate Value Property of continuous functions $g(x) - x$ has a root in $[a, b]$. Now suppose if possible, $\xi_1, \xi_2 \in (a, b)$ be such that $g(\xi_1) = \xi_1$ and $g(\xi_2) = \xi_2$. Then by mean value theorem there exists a point c in between ξ_1 and ξ_2 such that

$$g'(c) = \frac{g(\xi_2) - g(\xi_1)}{\xi_2 - \xi_1} = \frac{\xi_2 - \xi_1}{\xi_2 - \xi_1} = 1,$$

a contradiction to the assumption. Hence there is only one fixed point say ξ in (a, b) .

We will use Mathematical Induction to prove that the set $\{x_n : n \in \mathbb{N}\}$ is a subset of (a, b) . Clearly $x_2 = g(x_1) \in (a, b)$ since $x_1 \in (a, b)$. Next if $x_k \in (a, b)$, $x_{k+1} = g(x_k) \in (a, b)$. Note that

$$|x_{k+1} - x_k| = |g(x_k) - g(x_{k-1})| = |g'(\lambda_k)(x_k - x_{k-1})| = |g'(\lambda_k)| |x_k - x_{k-1}| \leq \alpha |x_k - x_{k-1}|.$$

Now since $\alpha < 1$, it can be proved that $\{x_k\}$ is a Cauchy sequence and hence convergent. If $x_k \rightarrow x$, then by continuity of g , $g(x_k) \rightarrow g(x)$ (Note that $x \in [a, b]$). And hence taking the limit in (5.19), we get $x = g(x)$. Thus x is the fixed point of g in (a, b) , or $x = \xi$. $\square$

Theorem 5.2. Let l_0 be a fixed point of $g(x)$. Suppose $\epsilon > 0$ be such that g is differentiable on $[l_0 - \epsilon, l_0 + \epsilon]$ and $|g'(x)| \leq \alpha < 1$ for all $x \in [l_0 - \epsilon, l_0 + \epsilon]$. Then the sequence defined by $x_n = g(x_{n-1})$ and $x_1 \in [l_0 - \epsilon, l_0 + \epsilon]$, converges to l_0 .

Proof. To prove this theorem we first prove that $g([l_0 - \epsilon, l_0 + \epsilon]) \subset [l_0 - \epsilon, l_0 + \epsilon]$. If $x \in [l_0 - \epsilon, l_0 + \epsilon]$, then $|l_0 - g(x)| = |g(l_0) - g(x)| = |g'(c)| |l_0 - x| \leq \alpha |l_0 - x| \leq \alpha \epsilon < \epsilon$, hence $g(x) \in [l_0 - \epsilon, l_0 + \epsilon]$. Now g fulfils the condition of Theorem (5.1) with interval $[a, b]$ as $[l_0 - \epsilon, l_0 + \epsilon]$ so we conclude the rest. $\square$

Remark 5.4. Suppose if g is twice differentiable, and $g'(r) \neq 0$, where r is the fixed point of g , then by Taylor's formula

$$x_{k+1} = g(x_k) = g(r + x_k - r) = g(r) + g'(r)(x_k - r) + \frac{1}{2}g''(c)(x_k - r)^2 = r + g'(r)e_k + \frac{1}{2}g''(c)e_k^2.$$

Thus,

$$\frac{|e_{k+1}|}{|e_k|} = |g'(r) + \frac{1}{2}g''(c)e_k|.$$

Now if $\{x_k\}$ converges to r or equivalently $e_k \rightarrow 0$, we have

$$\lim_{k \rightarrow \infty} \frac{|e_{k+1}|}{|e_k|} = |g'(r)|.$$

This shows that the **order of convergence of fixed point iteration method is 1** and asymptotic error is $|g'(r)|$.

Remark 5.5. Note that the iterations of Newton's method are given by

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}.$$

If $g(x) = x - \frac{f(x)}{f'(x)}$, then Newton's method is also a fixed point iteration method. From the last remark order of convergence of fixed point iteration method is 1, but we know that the **order of convergence of Newton's method is 2**. This ambiguity arises because in Newton's method $g'(r) = 0$, which is not the case in general for fixed point iteration method. Now let us examine the convergence behavior of the fixed point iteration method if $g'(r) = 0$ and $g''(r) \neq 0$. Suppose g is thrice differentiable, then by Taylor's formula

$$\begin{aligned} x_{k+1} &= g(x_k) = g(r + x_k - r) = g(r) + g'(r)(x_k - r) + \frac{1}{2}g''(r)(x_k - r)^2 + \frac{1}{6}g'''(c)(x_k - r)^3, \\ &= r + \frac{1}{2}g''(r)e_k^2 + \frac{1}{6}g'''(c)e_k^3. \end{aligned}$$

Thus

$$\frac{|e_{k+1}|}{|e_k|^2} = \left| \frac{1}{2}g''(r) + \frac{1}{6}g'''(c)e_k \right|, \quad \text{or} \quad \lim_{k \rightarrow \infty} \frac{|e_{k+1}|}{|e_k|^2} = \left| \frac{1}{2}g''(r) \right|.$$

Hence the convergence is of order 2 with asymptotic error as $|\frac{1}{2}g''(r)|$. Since for Newton's method $g(x) = x - \frac{f(x)}{f'(x)}$, the asymptotic error $|\frac{1}{2}g''(r)| = \frac{1}{2} \frac{|f''(r)|}{|f'(r)|}$.

5.10. Multiple Roots

If $f(x) = 0$ has a root at $x = r$ with multiplicity m greater than 1, then f can be written as $f(x) = (x - r)^m h(x)$, where h is a function such that $h(r)$ is nonzero finite quantity. Now

$$g(x) = x - \frac{(x - r)^m h(x)}{m(x - r)^{m-1} h(x) + (x - r)^m h'(x)} = x - \frac{x - r}{m} + \frac{(x - r)^2 h'(x)}{m^2 h(x) + m h'(x)(x - r)}.$$

And we have $g'(r) = 1 - \frac{1}{m}$. This shows that Newton's method converges linearly when $m > 1$. We aim to modify Newton's method to ensure the quadratic convergence. For this we modify g such that $g'(r) = 0$ and $g(r) = r$. Note that if we assume modified g as $g(x) = x - \alpha \frac{f(x)}{f'(x)}$, then $g'(r) = 1 - \frac{\alpha}{m}$. Thus we assume $g(x) = x - m \frac{f(x)}{f'(x)}$, so that modified sequence of iterates is

$$x_{k+1} = x_k - m \frac{f(x_k)}{f'(x_k)}. \quad (5.20)$$

Note that this sequence of iterates can be applied if we know a priori the multiplicity of the root. But to get the quadratic convergence, without knowing the multiplicity of the roots, we assume $g(x) = x - \frac{u(x)}{u'(x)}$, where $u(x) = \frac{f(x)}{f'(x)}$. Thus $g(r) = r$ and $g'(r) = 0$, and hence iterates defined by $x_{k+1} = g(x_k)$, converges quadratically irrespective of the multiplicity of the root of $f(x) = 0$. Thus in general we use following approximations to get quadratic convergence in case of roots with multiplicity greater than 1,

$$x_{k+1} = x_k - \frac{f(x_k)f'(x_k)}{(f'(x_k))^2 - f(x_k)f''(x_k)}. \quad (5.21)$$

Note that in this case we need an extra evaluation of $f''(x)$ in each iteration.

5.11. Solution of System of Nonlinear Equations

Consider the system of two nonlinear equations in two unknowns.

$$\begin{aligned} f(x, y) &= 0, \\ g(x, y) &= 0. \end{aligned}$$

Let $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by $F(x, y) = \begin{bmatrix} f(x, y) \\ g(x, y) \end{bmatrix}$. Using Taylor's series expansion for multi variable function we have

$$\begin{aligned} 0 &= f(x, y) = f(x_k + x - x_k, y_k + y - y_k) = f(x_k + \Delta x, y_k + \Delta y) \\ &= f(x_k, y_k) + [\Delta x_k(\partial/\partial x) + \Delta y_k(\partial/\partial y)]f|_{x_k, y_k} + \frac{1}{2}[\Delta x_k(\partial/\partial x) + \Delta y_k(\partial/\partial y)]^2 f|_{x_k, y_k} + \dots \end{aligned}$$

Neglecting higher order terms $[\Delta x_k(\partial/\partial x) + \Delta y_k(\partial/\partial y)]f|_{x_k, y_k} \approx -f(x_k, y_k)$. Thus we have

$$[\Delta x_k f_x + \Delta y_k f_y]|_{(x_k, y_k)} \approx -f(x_k, y_k),$$

$$[\Delta x_k g_x + \Delta y_k g_y]|_{(x_k, y_k)} \approx -g(x_k, y_k).$$

Or,

$$\begin{bmatrix} f_x & f_y \\ g_x & g_y \end{bmatrix}_{(x_k, y_k)} \begin{bmatrix} \Delta x_k \\ \Delta y_k \end{bmatrix} \approx - \begin{bmatrix} f \\ g \end{bmatrix}_{(x_k, y_k)}. \quad (5.22)$$

Let J denotes the Jacobian $\begin{bmatrix} f_x & f_y \\ g_x & g_y \end{bmatrix}$ and J_k denotes the Jacobian J evaluated at $X_k = [x_k, y_k]^t$. Thus we can write $J_k \Delta X_k \approx -F(X_k)$, or $\Delta X_k \approx -J_k^{-1} F(X_k)$, or $X \approx X_k - J_k^{-1} F(X_k)$, or

$$\begin{bmatrix} x \\ y \end{bmatrix} \approx \begin{bmatrix} x_k \\ y_k \end{bmatrix} - J_k^{-1} \begin{bmatrix} f(x_k, y_k) \\ g(x_k, y_k) \end{bmatrix}. \quad (5.23)$$

Thus we define sequence of iterates as

$$\begin{bmatrix} x_{k+1} \\ y_{k+1} \end{bmatrix} = \begin{bmatrix} x_k \\ y_k \end{bmatrix} - J_k^{-1} \begin{bmatrix} f(x_k, y_k) \\ g(x_k, y_k) \end{bmatrix}. \quad (5.24)$$

This method is known as Newton's iteration method of solving system of nonlinear equations. In general for system of n linear equations in n unknowns we define the iterates as

$$X_{k+1} = X_k - J_k^{-1} F(X_k), \quad (5.25)$$

where $X_k = [x_1^{(k)}, x_2^{(k)}, \dots, x_n^{(k)}]^t$, $F(X) = [f_1(X), f_2(X), \dots, f_n(X)]^t$ and J_k is the Jacobian of $f_1, f_2, \dots, f_n$ with respect to $x_1, x_2, \dots, x_n$ evaluated at $X_k = [x_1^{(k)}, x_2^{(k)}, \dots, x_n^{(k)}]^t$.

Remark 5.6. Suppose we have system of two linear equations say

$$ax + by = s \quad (5.26)$$

$$cx + dy = t. \quad (5.27)$$

We can view them as $f_1(x, y) = ax + by - s = 0$ and $f_2(x, y) = cx + dy - t = 0$. So that Jacobian J of f_1, f_2 with respect to x, y is $\begin{bmatrix} f_x & f_y \\ g_x & g_y \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, which is constant. Now we assume that $[x_0, y_0]^t$ is the initial approximation to the solution and J is invertible so that we can apply Newton's method to find first approximation to the solution say $[x_1, y_1]^t$ as

$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} - \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} \begin{bmatrix} f_1(x_0, y_0) \\ g(x_0, y_0) \end{bmatrix}. \quad (5.28)$$

Or,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} - \begin{bmatrix} f_1(x_0, y_0) \\ g(x_0, y_0) \end{bmatrix} = \begin{bmatrix} ax_0 + by_0 \\ cx_0 + dy_0 \end{bmatrix} - \begin{bmatrix} ax_0 + by_0 - s \\ cx_0 + dy_0 - t \end{bmatrix} = \begin{bmatrix} s \\ t \end{bmatrix}. \quad (5.29)$$

This shows that when we apply Newton's method to linear equations, then defining sequence of iterates becomes exact.